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Adaptive, Distribution-Free Uncertainty Quantification for Shear and Stoneley Wave Prediction Using Conformalized Ensemble Learning

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Abstract

The acquisition of shear (DTS) and Stoneley (DTST) wave transit times is indispensable for comprehensive reservoir characterization, yet it is often constrained by significant economic and operational limitations. This data gap has spurred the development of data-driven prediction models; however, a critical "credibility gap" persists. While modern machine learning (ML) models routinely achieve high point-accuracy metrics such as the coefficient of determination (R^2), these metrics alone do not guarantee geologically plausible or reliable predictions for high-stakes engineering decisions, including wellbore stability analysis and hydraulic fracture design. This paper directly addresses this credibility gap by introducing and validating a robust and reliable workflow that moves beyond deterministic point estimates to provide rigorous, probabilistic forecasts. The proposed methodology couples high-performance, tree-based ensemble regressors with Conformalized Quantile Regression (CQR), a framework that generates statistically valid, sample-specific prediction intervals without making restrictive distributional assumptions about the data or model errors. Validated using a comprehensive dataset from the geologically complex, heterogeneous carbonate reservoirs of the South Pars gas field, the study reveals nuanced model performance: Random Forest proves optimal for single-well prediction ($R^2 > 0.94$), whereas a Stacked Ensemble demonstrates superior robustness for inter-well generalization ($R^2 > 0.92$). Crucially, the CQR framework successfully generated adaptive 90% prediction intervals that achieved the target empirical coverage, confirming the method's validity. This work provides a practical, validated methodology to generate trustworthy, probabilistic forecasts of acoustic logs, enabling quantitative risk assessment and bridging the divide between high-accuracy ML and the reliability demands of modern geomechanical, petrophysical, and geophysical analysis and reservoir and drilling engineering.

The Geoscientific Imperative and the Credibility Gap in Acoustic Velocity Synthesis

The Foundational Role of Shear and Stoneley Waves in Reservoir Characterization

The accurate characterization of subsurface reservoirs is a fundamental prerequisite for the efficient and safe exploration, development, and management of hydrocarbon, geothermal, and carbon sequestration resources. Central to this endeavor is the measurement and interpretation of acoustic wave velocities, specifically compressional (V_p), shear (V_s), and Stoneley (V_{st}) waves. While compressional wave data are routinely acquired, the acquisition of shear and Stoneley wave data, often represented by their respective transit times (DTS and DTST), is frequently sparse or entirely absent due to significant economic and operational constraints. This data gap presents a formidable challenge, as both V_s and V_{st} provide unique and indispensable information for a comprehensive understanding of a reservoir's geomechanical, petrophysical, and geophysical properties. (Amerian *et al.*, 2025)

From a geomechanical perspective, shear wave velocity is the cornerstone for quantifying a rock's resistance to shear deformation. When combined with compressional wave velocity and bulk density (ρ_b), V_s is essential for calculating the dynamic elastic moduli that governs rock behavior under stress, including Shear Modulus (μ), Young's Modulus (E), and Poisson's Ratio (ν). These parameters are critical inputs for high-stakes engineering applications such as defining the safe mud weight window for drilling, designing effective hydraulic fracturing programs, and estimating in-situ stress states. An error in a predicted V_s value propagates non-linearly through these calculations, potentially amplifying uncertainty and leading to flawed engineering designs with catastrophic consequences. Petrophysically, the V_p/V_s ratio is a powerful diagnostic tool for discriminating fluid types and differentiating lithologies. In geophysics, V_s is a vital input for Amplitude-Versus-Offset (AVO) analysis and pre-stack seismic inversion. The Stoneley wave offers a unique window into formation permeability, as its velocity and attenuation are highly sensitive to fluid mobility in the near-wellbore region. Consequently, the development of robust methodologies to predict these missing logs from commonly available data is a critical enabler of modern reservoir engineering workflows. (Hazbeh *et al.*, 2024)

A Critical Appraisal of Conventional and Data-Driven Prediction Paradigms

The challenge of estimating shear wave velocity has been addressed through an evolution of methodologies. The earliest approaches involved simple empirical correlations, such as the linear "mud-rock line" by Castagna *et al.* (Castagna, Batzle and Eastwood, 1985), which relate V_s to V_p . While simple, these models are intrinsically tied to the specific geological context of their derivation data and exhibit a profound lack of transferability, failing significantly when applied to different formations. Rock physics models, such as the Xu-White (Xu and White, 1995) and Xu-Payne (Xu and Payne, 2009) models, offered a more theoretical framework by linking elastic properties to fundamental rock constituents. However, their practical application is hindered by a high sensitivity to unmeasurable parameters like pore aspect ratio, which is often used as a non-unique calibration factor, especially in geologically complex carbonates.

These limitations catalyzed a definitive shift towards data-driven paradigms. Machine learning (ML) and, more recently, deep learning (DL) models are capable of learning complex, non-linear relationships directly from data without being constrained by rigid physical or statistical assumptions. (Alimohammadi, Mahmoudi and Chen, 2020) The application of these techniques has progressed from early "intelligent systems" like Fuzzy Logic to the mainstreaming of Artificial Neural Networks (ANNs) and Support Vector Regression (SVR). The current state-of-the-art for tabular data prediction is dominated by ensemble learning methods, particularly Random Forest (RF) and Gradient Boosting Machines (e.g., XGBoost, LightGBM), which consistently deliver the highest accuracy in comparative studies. More recent advances have leveraged deep learning architectures like Long Short-Term Memory (LSTM) networks and 1D

Convolutional Neural Networks (1D-CNNs) to explicitly model the sequential nature of well log data, leading to further performance gains, particularly in hybrid CNN-LSTM models. A summary of key developments of different approaches is provided in [Table 1](#).

Table 1—A Chronological Summary of Key Machine Learning and Deep Learning Studies for Acoustic Velocity Synthesis

Reference (Year)	Methodology	Key Input Features	Reservoir Type / Lithology	Reported Performance (Metric, Value)	Key Finding / Contribution
Rajabi et al. (2010)(Rajabi, Bohloli and Gholampour Ahangar, 2010)	FL, GA, NF (ANFIS)	LLD, RHOB, NPHI	Carbonate (Sarvak Fm.)	MSE = 0.0084 (Vs), $R^2 \approx 0.95$ (Vs)	Fuzzy Logic (FL) performed best; proved applicability in complex carbonates.
Labani & Sabzekar (2023)(Labani and Sabzekar, 2023)	CMIS (NN, ANFIS) + GA	NPHI, RHOZ, Vp	Carbonate (South Pars)	$R^2 = 0.9364$ (Vs), $R^2 = 0.9270$ (Vst)	Committee machines improve on individual models; introduced Monte-Carlo UQ.
Zhang et al. (2020)(Zhang et al., 2020)	LSTM	AC, GR, PE, DEN, RLLD, CNL	Carbonate (Sulige Gas Field)	Accuracy=98.9% (Vs. 73% for Xu-Payne)	LSTM captures sequential nature of logs, significantly outperforming rock physics.
Zhang et al. (2022)(Zhang et al., 2022)	1D-CNN	Conventional, Array Induction, Spectral GR logs	Sandstone & Carbonate (Ordos Basin)	$R^2 = 0.979$ (Sandstone), $R^2 = 0.980$ (Carbonate)	1D-CNN effectively extracts features from log sequences, outperforming traditional ML.
Hazbeh et al. (2024)(Hazbeh et al., 2024)	DBN, RF	MD, CALI, GR, Resistivity, DCT, NPHI, PEF	SW Iran Oil Field	$R^2 = 0.9775$ (DBN) Vs 0.8669 (RF)	Deep Belief Network (DBN) excels in deep feature extraction and outperforms RF.
Mollaei et al. (2025) (Mollaei, Moradzadeh and Mohebian, no date)	MLP, LSTM, CNN, CNN+LSTM	CHAL, RHOB, Vp, GR, NEUT	Southern Iran Hydrocarbon Field	$R^2 = 0.9934$ (CNN +LSTM on blind data)	Combined CNN +LSTM is the most accurate, confirming hybrid DL superiority.
Amerian et al. (2025)(Amerian et al., 2025)	RF, GB, CatBoost, etc.	CGR, RHOB, DTC, Lithology	Mixed Lithologies	Test $R^2 > 0.97$ (RF for DTS)	Confirms ensemble dominance and warns that high R^2 can be misleading.

The Frontier: From the Commoditization of Accuracy to the Mandate for Uncertainty Quantification

The consistent outperformance of modern ML/DL models has led to a "commoditization of accuracy," where achieving a high coefficient of determination (R^2) is no longer a novel research contribution but an expected baseline. This focus on single-point accuracy metrics has revealed a critical "credibility gap." As Amerian et al. (Amerian et al., 2025) emphatically caution, high R^2 or low RMSE alone "do not guarantee valid predictions" and can be "misleading in practice". They demonstrated that even models with an R^2 approaching 0.99 can produce geologically implausible results, such as unrealistic Poisson's ratios, when the predicted velocities are used in subsequent calculations. A model can learn to be a highly sophisticated interpolation engine within its training data distribution but may fail to respect the underlying physical constraints of rock physics when forced to extrapolate, even slightly.

This creates a situation where a single, deterministic prediction, regardless of its supposed accuracy, provides no information about its own reliability or the potential range of errors. For any high-stakes engineering decision, knowing the most likely value is insufficient; one must also know the range of plausible values to properly assess risk. This has created an emerging and high uncertainty Quantification

(UQ) in petrophysical prediction. By moving beyond a single point estimate to provide a probabilistic forecast or a prediction interval, UQ gives decision-makers a quantitative measure of the confidence they can place in a model's output.

While the need for UQ is clear, a significant research gap remains for a framework that can provide uncertainty bounds that are simultaneously rigorous, calibrated, and sample-specific. Most traditional error metrics, like a global RMSE, provide only an average measure of error across an entire dataset and do not indicate whether a specific prediction for a given depth point is likely to be highly accurate or highly uncertain. (Alimohammadi, Rahmanifard and Chen, 2022; Alimohammadi and Chen, 2023) What is required for practical engineering application is a framework that can deliver:

1. **Rigorous Guarantees:** Possesses strong theoretical, statistical guarantees of performance.
2. **Distribution-Free:** Makes no assumptions about the underlying distribution of the data or the model's errors.
3. **Calibrated Performance:** The generated uncertainty bounds are empirically reliable; for example, a 90% prediction interval should contain the true value 90% of the time.
4. **Sample-Specific Granularity:** Provides a unique uncertainty estimate for each individual prediction point, reflecting local data quality and geological complexity.
5. **Model-Agnostic Flexibility:** Can be wrapped around any underlying predictive model (e.g., RF, XGBoost, LSTM).

This precisely defined gap creates the intellectual launchpad for this work, which introduces a framework meeting these criteria to provide a practical and trustworthy tool for reliable shear and Stoneley waves prediction and quantitative reservoir characterization.

A Framework for Rigorous, Model-Agnostic Uncertainty Quantification: Conformal Prediction

To address the critical need for reliable uncertainty quantification (UQ) in multivariate regression analysis, this study employs the framework of Conformal Prediction (CP). CP is a set of techniques that can augment any point prediction algorithm, from simple linear regression to complex deep neural networks, to produce prediction regions (e.g., intervals) with explicit, non-asymptotic, and distribution-free guarantees on their coverage probability (Oliveira *et al.*, no date). This framework is particularly valuable in applications like reservoir characterization, where data heterogeneity and uncertainty are prevalent, as highlighted in the prediction of shear (DTS) and Stoneley (DTST) wave transit times.

Theoretical Foundations: Exchangeability, Nonconformity, and Guaranteed Coverage

The theoretical strength of conformal prediction lies in its reliance on a single, relatively weak assumption: exchangeability (Oliveira *et al.*, no date). A sequence of random variables is exchangeable if its joint probability distribution is invariant to any permutation of the sequence. While the common assumption of independent and identically distributed (i.i.d.) data implies exchangeability, the reverse is not true, making exchangeability a weaker and more general condition (Hutter, 2022). Under this assumption, CP provides a frequentist guarantee of coverage: for a user-specified mis-coverage rate $\alpha \in (0, 1)$, the generated prediction interval $C(X_{n+1})$ for a new data point X_{n+1} will contain the true outcome Y_{n+1} with a probability of at least $(1 - \alpha)$.

The core mechanism of CP is the nonconformity score, a function $s(x, y)$ that measures how "strange" or "atypical" a given example (x, y) is with respect to the underlying predictive model, f (Vovk *et al.*, 2020). A higher score indicates that the example is more non-conforming, meaning the model's prediction $\hat{y} = f(x)$ is a poor fit for the true label y . For regression tasks, the most common nonconformity score is the absolute residual (Clarkson *et al.*, 2024):

$$s(x, y) = |y - \hat{y}|$$

This score quantifies the magnitude of the prediction error.

The framework uses these scores to compute a p-value for any hypothetical new label, y_{test} , for a new test point, x_{test} . This p-value is defined as the fraction of nonconformity scores on a set of past calibration examples that are greater than or equal to the score of the test example with the hypothetical label (Shafer, Vovk and Uk, no date):

$$p(x_{\text{test}}, y_{\text{test}}) = \frac{|\{i = 1, \dots, n_{\text{cal}} : s(x_i, y_i) \geq s(x_{\text{test}}, y_{\text{test}})\}| + 1}{n_{\text{cal}} + 1}$$

where $s_i = s(x_i, y_i)$ for $i = 1, \dots, n_{\text{cal}}$ is the set of calibration examples. The prediction interval $C(x_{\text{test}})$ is then constructed as the set of all hypothetical labels y_{test} whose p-value is greater than the specified mis-coverage rate α (Jin and Ren, 2025).

Practical Implementation: The Split Conformal Prediction (SCP) Workflow

The original "transductive" form of CP requires retraining the underlying model for each hypothetical label, making it computationally prohibitive for most modern machine learning algorithms. The most widely used practical alternative is Inductive Conformal Prediction (ICP), also known as Split Conformal Prediction (SCP). SCP achieves computational efficiency by splitting the available data once and training the model only once (Han *et al.*, 2022; Sousa, 2022).

The SCP workflow is as follows:

1. **Data Split:** The available labeled data is randomly partitioned into two disjoint sets: a proper training set, D_{train} , and a calibration set, D_{cal} .
2. **Model Training:** An arbitrary point prediction model, f , is trained using only the proper training set, D_{train} .
3. **Calibration:** The trained model $\hat{f}$ is used to make predictions on the calibration set. The nonconformity scores $s_i = s(x_i, y_i)$ are computed for all $(x_i, y_i) \in D_{\text{cal}}$. This yields a set of n_{cal} scores, $s_1, s_2, \dots, s_{n_{\text{cal}}}$.
4. **Quantile Determination:** The $(1 - \alpha)(1 + 1/n_{\text{cal}})$ quantile of the calibration scores, denoted $\hat{q}$, is calculated. This value represents the critical threshold of nonconformity.
5. **Prediction Interval Construction:** For a new, unseen test point x_{new} , a point prediction $\hat{y}_{\text{new}} = \hat{f}(x_{\text{new}})$ is made. The final prediction interval is constructed by adding and subtracting the threshold $\hat{q}$ from the point prediction:

$$C(x_{\text{new}}) = [\hat{y}_{\text{new}} - \hat{q}, \hat{y}_{\text{new}} + \hat{q}]$$

6. This procedure ensures that, on average, no more than a fraction α of the true test values will fall outside their respective prediction intervals, provided the data are exchangeable (Angelopoulos, Barber and Bates, 2024).

However, a significant limitation of SCP is that it produces prediction intervals of fixed width for all test points. This is suboptimal for applications like reservoir characterization, where data uncertainty is often heteroscedastic, i.e., the variability of the target variable changes with the input features. In such cases, predictions in simple, uniform lithologies should be more certain (narrower intervals) than those in complex, diagenetically altered zones (wider intervals).

Advanced Methods for Heterogeneous Reservoirs: Conformalized Quantile Regression (CQR) for Adaptive, Sample-Specific Intervals

To address the limitations of standard conformal prediction methods like SCP, Conformalized Quantile Regression (CQR) has been introduced as a more advanced approach. CQR combines the statistical

efficiency of quantile regression with the rigorous coverage guarantees of conformal prediction, producing adaptive, sample-specific prediction intervals that account for heteroscedasticity. This is particularly relevant for heterogeneous carbonate reservoirs, where variability in acoustic log predictions (e.g., DTS and DTST) depends on geological features.

The CQR workflow modifies the standard SCP procedure as follows (Sesia and Candès, 2020):

1. **Data Split:** The data is split into a proper training set, D_{train} , and a calibration set, D_{cal} , as before.
2. **Quantile Model Training:** Instead of a single point predictor, two quantile regression models are trained on D_{train} . These models estimate the conditional lower and upper quantiles of the target variable, typically $\hat{q}_{\text{low}}(x)$ for the $\alpha/2$ quantile and $\hat{q}_{\text{high}}(x)$ for the $1 - \alpha/2$ quantile. These models can be any algorithm supporting quantile regression, such as Gradient Boosting Machines with a quantile loss function or Random Forests adapted for quantile regression.
3. **Adaptive Nonconformity Score:** For each example (x_i, y_i) in the calibration set D_{cal} the score E_i is computed as the signed distance from the true value y_i to the nearest boundary of the predicted quantile interval:

$$E_i = \max(\hat{q}_{\text{low}}(x_i) - y_i, y_i - \hat{q}_{\text{high}}(x_i))$$

This score is positive if the true value falls outside the predicted quantile interval and non-positive otherwise.

4. **Quantile Determination:** The $(1 - \alpha)$ quantile of these scores, denoted $\hat{q}^{cqr}$, is calculated from the empirical distribution of the E_i scores.
5. **Adaptive Prediction Interval Construction:** For a new test point x_{new} , the quantile models predict $\hat{q}_{\text{low}}(x_{\text{new}})$ and $\hat{q}_{\text{high}}(x_{\text{new}})$. The final CQR prediction interval is:

$$C(x_{\text{new}}) = [\hat{q}_{\text{low}}(x_{\text{new}}) - \hat{q}^{cqr}, \hat{q}_{\text{high}}(x_{\text{new}}) + \hat{q}^{cqr}]$$

The key advantage of CQR is that the initial interval width, $(\hat{q}_{\text{high}}(x) - \hat{q}_{\text{low}}(x))$, adapts to the input features x , reflecting local uncertainty. The conformalization step then ensures a uniform coverage guarantee of $1 - \alpha$. This makes CQR particularly suitable for heterogeneous reservoirs, where uncertainty varies across different lithologies or geological formations. Research suggests that CQR produces shorter, more efficient intervals compared to standard conformal methods, as demonstrated in experiments on various regression datasets (Chernozhukov, Wüthrich and Zhu, 2021).

Application to Multivariate Regression in Reservoir Characterization

In the context of multivariate regression analysis for reservoir characterization, CQR provides a powerful tool for generating probabilistic forecasts of acoustic logs (e.g., shear wave transit times, DTS, and Stoneley wave transit times, DTST). These forecasts must account for the complex, heterogeneous nature of carbonate reservoirs, where uncertainty is not uniform across the input space. By leveraging CQR, we can produce prediction intervals that are both statistically rigorous and adaptive to local variability, ensuring reliable uncertainty quantification for high-stakes engineering decisions such as wellbore stability analysis and hydraulic fracture design. Recent advancements, such as ensemble CQR for time series or clustering-based improvements to CQR, further enhance its applicability to complex datasets (Jensen, Bianchi and Anfinson, 2024).

Comparison with Other Methods

CQR offers several advantages over traditional uncertainty quantification methods like standard quantile regression or Bayesian approaches. Unlike quantile regression alone, which may produce mis-calibrated intervals, CQR ensures valid coverage through its conformalization step. Compared to Bayesian methods,

CQR requires fewer assumptions about data distribution and is computationally more efficient, as it avoids the need for complex posterior sampling. Additionally, CQR's model-agnostic nature allows it to be paired with high-performance models like Random Forests or Gradient Boosting Machines, which are well-suited for the high-dimensional, nonlinear relationships in reservoir data. A summary of key differences of different methods is provided in [Table 2](#)(Dewolf, 2024).

Table 2—Comparison of Uncertainty Quantification Methods

Method	Coverage Guarantee	Handles Heteroscedasticity	Distributional Assumptions	Computational Efficiency
Standard Quantile Regression	No	Yes	Yes	High
Bayesian Regression	Yes (approximate)	Yes	Yes	Low
Split Conformal Prediction	Yes	No	No	High
Conformalized Quantile Regression	Yes	Yes	No	Moderate

Methodology: From Petrophysical Logs to Probabilistic Forecasts

The proposed workflow transforms standard petrophysical well log data into probabilistic forecasts for shear and Stoneley wave velocities. This process involves four main stages: data preparation, development of high-performance point prediction models, generation of uncertainty intervals using conformal prediction, and a comprehensive evaluation protocol.

Data Preparation and Feature Engineering

The dataset for this study comprises well log data from multiple wells drilled in the South Pars gas field. The data covers a significant depth range across the primary reservoir intervals. Before being used for model training, the raw log data underwent a standard preprocessing pipeline to ensure data quality and consistency. This included:

- **Missing Value Imputation:** Handling of gaps in log curves using a combination of linear interpolation for small gaps and multivariate imputation for larger intervals.
- **Outlier Detection and Clipping:** Identification and removal of unrealistic spikes or data points resulting from logging tool errors, using a moving window standard deviation filter.
- **Data Normalization:** Scaling of all input features to a common range (e.g., 0 to 1) using min-max scaling to ensure that features with larger numeric ranges do not disproportionately influence model training.

The selection of input features for predicting the target variables, Shear Transit Time (DTS) and Stoneley Transit Time (DTST), was guided by petrophysical principles and a review of established practices in the literature. The final feature set, summarized in [Table 3](#), includes a standard suite of logs that are sensitive to lithology, porosity, fluid content, and borehole conditions.

Table 3—Input Petrophysical Logs and Engineered Features

Mnemonic	Description	Unit	Role
DTC	Compressional Sonic Transit Time	μs/ft	Input Feature
GR	Gamma Ray	GAPI	Input Feature
RHOB	Bulk Density	g/cm³	Input Feature
NPHI	Neutron Porosity	v/v	Input Feature

Mnemonic	Description	Unit	Role
LLD	Laterolog Deep Resistivity	ohm.m	Input Feature
CALI	Caliper	in	Input Feature
PEF	Photoelectric Factor	barns/electron	Input Feature
DTS	Shear Sonic Transit Time	$\mu\text{s}/\text{ft}$	Target Variable
DTST	Stoneley Sonic Transit Time	$\mu\text{s}/\text{ft}$	Target Variable

Point Prediction Model Development: Training Multivariate Ensemble Regressors

Based on a comprehensive literature review demonstrating the strong performance of both regularized linear models and tree-based ensemble methods on a wide variety of tabular datasets, we selected six complementary algorithms to develop our point-prediction models (Shwartz-Ziv and Armon, 2022; Rana *et al.*, 2023; Uddin and Lu, 2024). First, we include three penalized regression techniques, Lasso, Ridge, and ElasticNet, which introduce L1 and/or L2 regularization to the ordinary least squares framework. These methods not only mitigate overfitting by shrinking less informative feature coefficients toward zero, but also, in the case of Lasso and ElasticNet, perform automatic feature selection, yielding more interpretable models when the underlying signal is sparse.

In addition, we employ four decision-tree-based ensemble learners, Random Forest (RF), XGBoost, LightGBM, and CatBoost, each of which combines multiple base trees to boost predictive accuracy and stability. Random Forest aggregates a diverse set of bootstrap-sampled trees via bagging, reducing variance and improving robustness against noisy data. XGBoost and LightGBM both implement gradient boosting, sequentially fitting trees to the residual errors of prior models; LightGBM further optimizes training speed and memory usage through histogram-based splitting and leaf-wise growth. Finally, CatBoost introduces ordered boosting and native handling of categorical features, often delivering superior performance with minimal preprocessing.

By pairing interpretable linear approaches with powerful, nonlinear ensembles, our modeling suite is designed to capture a broad spectrum of relationships, ranging from simple additive effects to highly complex interactions, while controlling overfitting and ensuring generalizability across diverse tabular domains.

For each target variable (DTS and DTST), separate linear and tree-based models were trained. The model development process involved:

- **Data Partitioning:** The full dataset was first split into a development set (90%) and a final, held-out test set (10%). The development set was then further partitioned into a proper training set (80% of development data) and a calibration set (20% of development data), following the split conformal prediction paradigm.
- **Hyperparameter Tuning:** A randomized search with 5-fold cross-validation was performed on the proper training set to identify the optimal hyperparameters for each model (e.g., number of trees, tree depth, learning rate). This ensures that the models are well-tuned without seeing the calibration or test data, preventing data leakage.
- **Model Training:** The final point prediction models were trained on the entire proper training set using the best-identified hyperparameters.

Uncertainty Interval Generation: Implementing plus Method

With the trained point predictors, the next step was to generate the uncertainty intervals using the calibration set. The plus method of the Python library MAPIE (version 0.7.1) (Taquet *et al.*, 2022), which provides a user-friendly and robust implementation of various conformal prediction methods, was used for this purpose. A target mis-coverage rate of $\alpha=0.1$ was selected, corresponding to a desired coverage level of 90%

for the prediction intervals. When configured with `method="plus"`, `MapieRegressor` constructs prediction intervals by combining two key ideas, cross-validated error assessment and an ensemble of fold-specific predictions, while making no distributional assumptions about model residuals. First, the training set is partitioned into K folds and, in turn, each fold is held out while the base learner is trained on the remaining data. The errors observed on these held-out points form an empirical distribution of the model's "typical mistakes." Next, each of the K models is used to predict a new observation, yielding an ensemble of K candidate predictions for that point. By adding each recorded error to each candidate prediction, one obtains a large collection of plausible outcomes. Finally, MAPIE simply selects the appropriate lower and upper percentiles of this combined set, matching the user's desired confidence level, to define the interval.

Evaluation Protocol: Metrics for Point Accuracy and Prediction Interval Validity

A dual-level evaluation protocol was established to assess both the accuracy of the point predictions and the quality of the uncertainty intervals.

- **Point Accuracy Metrics:** The performance of the underlying linear and tree-based models was evaluated on the held-out test set using two standard metrics:
 - **Coefficient of Determination (R^2):** Measures the proportion of the variance in the dependent variable that is predictable from the independent variables.
 - **Root Mean Squared Error (RMSE):** Measures the standard deviation of the prediction errors (residuals).
 - **Mean Absolute Error (MAE):** Measures the average magnitude of the prediction errors, without considering their direction. It provides a straightforward interpretation of the average error between predicted and actual values.
- **Prediction Interval Quality Metrics:** The performance of the prediction intervals was evaluated on the test set using two key metrics that capture validity and efficiency ¹⁴:
 - **Empirical Coverage:** The actual percentage of true response values in the test set fall within their corresponding prediction intervals. For a valid conformal predictor, this value should be very close to the target coverage of $1-\alpha$ (i.e., 90%).
 - **Average Interval Width:** The mean width of all prediction intervals generated for the test set. This metric quantifies the efficiency or informativeness of the intervals. A smaller average width indicates a more precise and useful prediction, provided that the empirical coverage is maintained.
 - **Coverage Efficiency:** A composite metric that balances empirical coverage and interval width. It reflects how effectively the method maintains the desired coverage while minimizing the width of the intervals. Higher coverage efficiency indicates better performance in terms of both reliability and precision.

Case Study: The South Pars Gas Field

To evaluate the proposed multivariate regression methodology for predicting shear and Stoneley wave velocities, this study utilizes well log data from three wells in the South Pars gas field. This field is renowned for its vast non-associated gas reserves and complex carbonate reservoir system. The significant geological heterogeneity and diagenetic overprinting of the reservoir make it an exemplary case for testing adaptive uncertainty quantification in challenging subsurface environments. (Aghchelow, Nabi-Bidhendi and Shahvar, 2012)

Geological Setting and Stratigraphy: The Permo-Triassic Kangan and Dalan Formations

The South Pars field is the Iranian portion of the world's largest non-associated gas accumulation, the South Pars/North Dome field, on the shared border between Iran and Qatar. The field is situated on the Qatar Arch, a major N-S trending structural feature. Gas accumulation is primarily hosted within the carbonate-evaporite series of the Upper Permian Dalan and the Lower Triassic Kangan formations, which are stratigraphic equivalents of the prolific Khuff Formation in the Arabian Peninsula. (Tavakoli, Rahimpour-Bonab and Esrafil-Dizaji, 2010a)

These formations are subdivided into four main reservoir zones, designated K1 to K4 from top to bottom. The Kangan Formation comprises the K1 and K2 zones, while the Dalan Formation contains the K3 and K4 zones. The primary producing intervals are the K2 and K4 zones, which are predominantly composed of limestone and dolomite, while the K1 and K3 zones contain more anhydrite and exhibit lower reservoir quality. These reservoir units are separated by impermeable anhydrite layers, with the entire sequence capped by the thick anhydrite and shale of the Triassic Dashtak Formation, which acts as an effective regional seal. The source rock for this supergiant gas accumulation is believed to be the organic-rich Lower Silurian shales, with gas having migrated laterally into the Qatar Arch structure. (Moghaddas *et al.*, 2017)

Reservoir Characterization: A Complex, Diagenetically-Altered Carbonate System

The primary challenge in characterizing the South Pars reservoir lies in its extreme heterogeneity, which is a product of both its depositional environment and a complex, multi-stage diagenetic history. The sediments were deposited in a shallow marine, low-angle homoclinal carbonate ramp setting. However, the original depositional fabric and primary porosity have been significantly modified by subsequent diagenetic overprinting. (Shirazinia *et al.*, 2020)

Key diagenetic processes that control the final reservoir quality include (Tavakoli, Rahimpour-Bonab and Esrafil-Dizaji, 2010b):

- **Dolomitization:** A widespread process that can either enhance reservoir quality by creating intercrystalline porosity or reduce it if dolomite crystals grow to occlude pore space. Fabric-destructive dolomitization is a major cause of high reservoir quality in the K2 and K4 zones
- **Dissolution:** Meteoric and burial dissolution have created significant secondary porosity in the form of molds and vugs, particularly in grain-dominated facies.
- **Cementation:** Pervasive calcite and anhydrite cementation acts as a primary control on reservoir quality destruction. Anhydrite, in particular, can be found as pore-filling, nodular, and bedded fabrics that drastically reduce porosity and permeability.
- **Compaction:** Both mechanical and chemical compaction (stylolitization) have reduced primary porosity, especially in mud-dominated facies.

This interplay between depositional textures and intense diagenetic alteration results in a highly complex pore system and makes the relationship between petrophysical log responses and rock properties (including acoustic velocities) exceptionally non-linear and spatially variable. This establishes the South Pars field as a rigorous proving ground for any advanced prediction methodology. A summary of generalized geological and reservoir properties of the South Pars field is provided in Table 4.

Table 4—Generalized Geological and Reservoir Properties of the South Pars Field (Hatampour, Schaffie and Jafari, 2015)

Formation	Member/Zone	Age	Dominant Lithology	Reservoir Quality
Kangan	K1	Early Triassic	Dolomite, Anhydrite, Limestone	Secondary Reservoir / Seal
	K2	Early Triassic	Limestone, Dolomite	Primary Reservoir
Dalan	K3	Late Permian	Dolomite, Anhydrite, Limestone	Secondary Reservoir / Seal
	K4	Late Permian	Limestone, Dolomite	Primary Reservoir

Pore System Complexity. The reservoir's pore system reflects a blend of primary depositional features and secondary diagenetic modifications. Primary interparticle and intraparticle porosity is frequently overprinted by dissolution and cementation. In the K2 and K4 zones, moldic and vuggy porosity dominates, acting as scattering centers for acoustic waves and complicating velocity predictions. Microporosity within dolomite crystals adds further intricacy, as it influences log responses but is difficult to quantify without high-resolution data.

Fracture Networks and Their Impact. Although not ubiquitous, natural fractures occur in brittle dolomitic intervals, particularly within the K2 and K4 zones. These fractures enhance permeability but introduce anisotropy in acoustic wave propagation, challenging isotropic models. While this study prioritizes matrix properties, fracture effects on DTS and DTST must be considered in high-fracture-density zones, potentially requiring model adjustments.

Reservoir Quality Variability. Reservoir quality varies significantly across the K1 to K4 zones. The K2 and K4 zones, with their dolomite- and limestone-dominated lithologies, exhibit higher porosity (up to 20%) and permeability (10–100 mD), making them the primary targets. Conversely, the anhydrite-rich K1 and K3 zones have lower porosity (<5%) and act as baffles or seals. This vertical and lateral variability necessitates detailed log analysis to inform regression models effectively.

The Petrophysical Challenge: Data and Heterogeneity in the Study Wells

The wells selected for this study penetrate the full Kangan and Dalan sequences, providing a comprehensive dataset that captures the vertical heterogeneity of the reservoir. The available log suite, while standard, presents a classic prediction challenge: to infer the complex geomechanical properties encoded in DTS and DTST from logs that are primarily sensitive to lithology, porosity, and fluid type. Previous studies in the South Pars field have successfully applied various intelligent systems, from fuzzy logic to committee machines, to predict petrophysical properties, establishing a precedent for the use of data-driven methods in this complex environment. This work builds upon that foundation by not only providing an accurate prediction but, for the first time, rigorously quantifying its uncertainty in a sample-specific manner, directly addressing the challenges posed by the field's geological complexity (Gharechelou *et al.*, 2022).

The open-hole logs from Well A (Figure 1), including Caliper, CGR (Computed Gamma Ray), DTC (Compressional Wave Slowness), PEF (Photoelectric Factor), RHOB (Bulk Density), RT (True Resistivity), RXO (Shallow Resistivity), and TNPH (Thermal Neutron Porosity), serve as critical inputs for the multivariate regression analysis aimed at predicting the shear wave slowness (DTS). The non-linear relationships between these logs and the target log, DTS, highlight the need for a sophisticated modeling approach. For instance, the interplay of porosity (TNPH) and density (RHOB) with acoustic slowness reflects the combined influence of lithology and diagenesis, while logs like CGR and PEF provide insights into mineral composition affecting shear wave velocities. Additionally, DTC offers a direct empirical link to DTS, though this relationship varies in complex reservoirs like South Pars. By integrating these diverse measurements, the regression methodology captures these complexities, and the sample-specific uncertainty quantification provides a reliability metric for each prediction, essential for decision-making in such a heterogeneous reservoir.

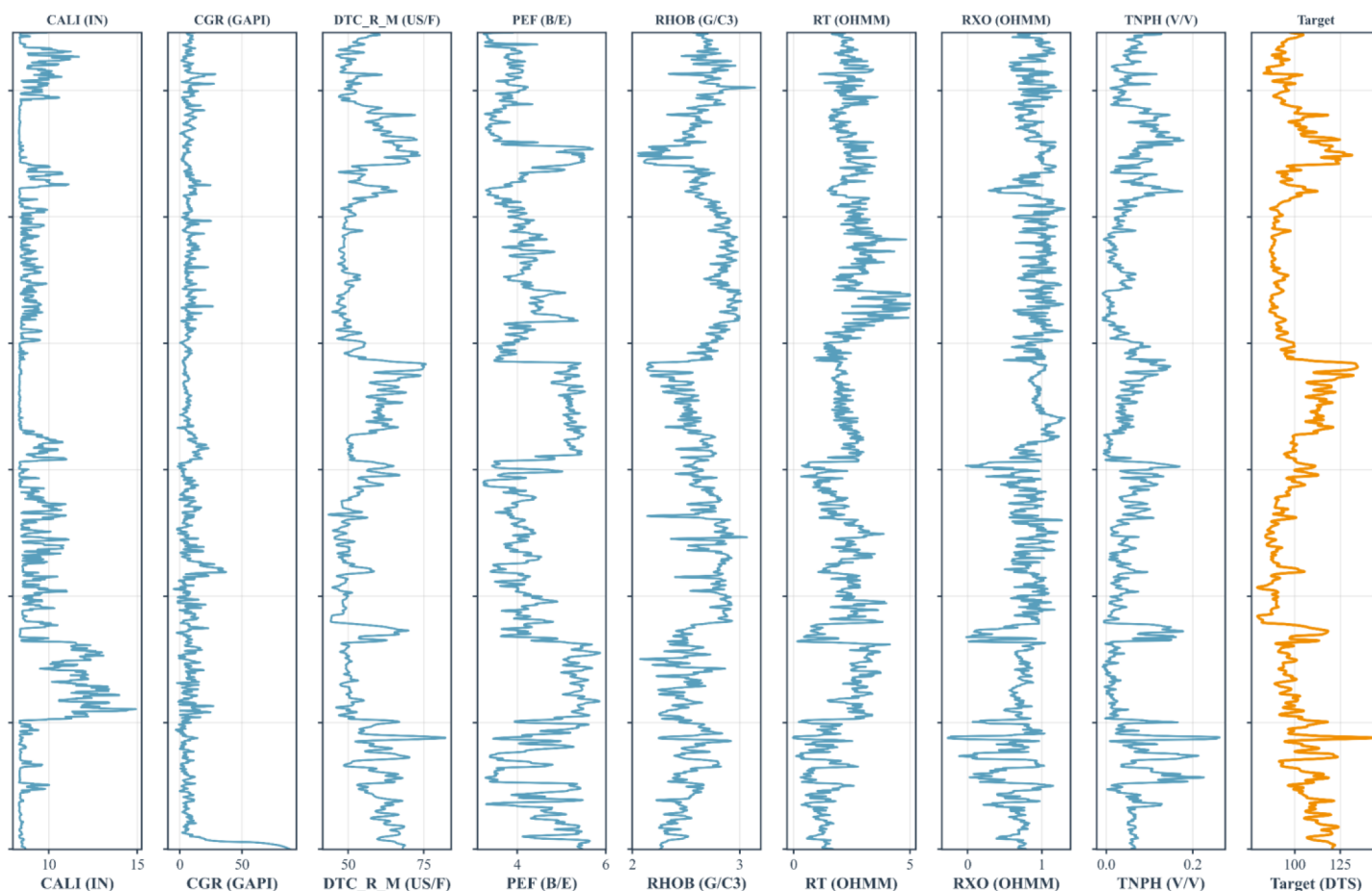


Figure 1—Open-hole logs from one of Well A. This figure displays the Caliper, CGR, DTC, PEF, RHOB, RT, RXO, TNPH, and DTS logs, providing a comprehensive view of the well's petrophysical properties across the K1 to K4 zones.

Figure 2 presents the distribution of each open-hole log feature used in the multivariate regression analysis. For each feature, a histogram is displayed alongside a Kernel Density Estimate (KDE) curve, which provides a smoothed representation of the data's probability density function. The mean value of each feature is also indicated, serving as a measure of central tendency. This visualization is essential because it reveals key statistical properties of each log, such as skewness, kurtosis, and the presence of outliers, which can significantly influence the performance of regression models. For example, features with non-normal distributions may require transformations (e.g., logarithmic or square root) to stabilize variance and enhance model accuracy. (Alimohammadi and Nancy Chen, 2022) Additionally, understanding the distribution of the target variable, DTS, is crucial for evaluating the suitability of the chosen regression technique and interpreting the model's predictive capabilities within the heterogeneous South Pars reservoir. By examining these distributions, potential data issues can be identified early, ensuring that the regression model is built on a solid statistical foundation.

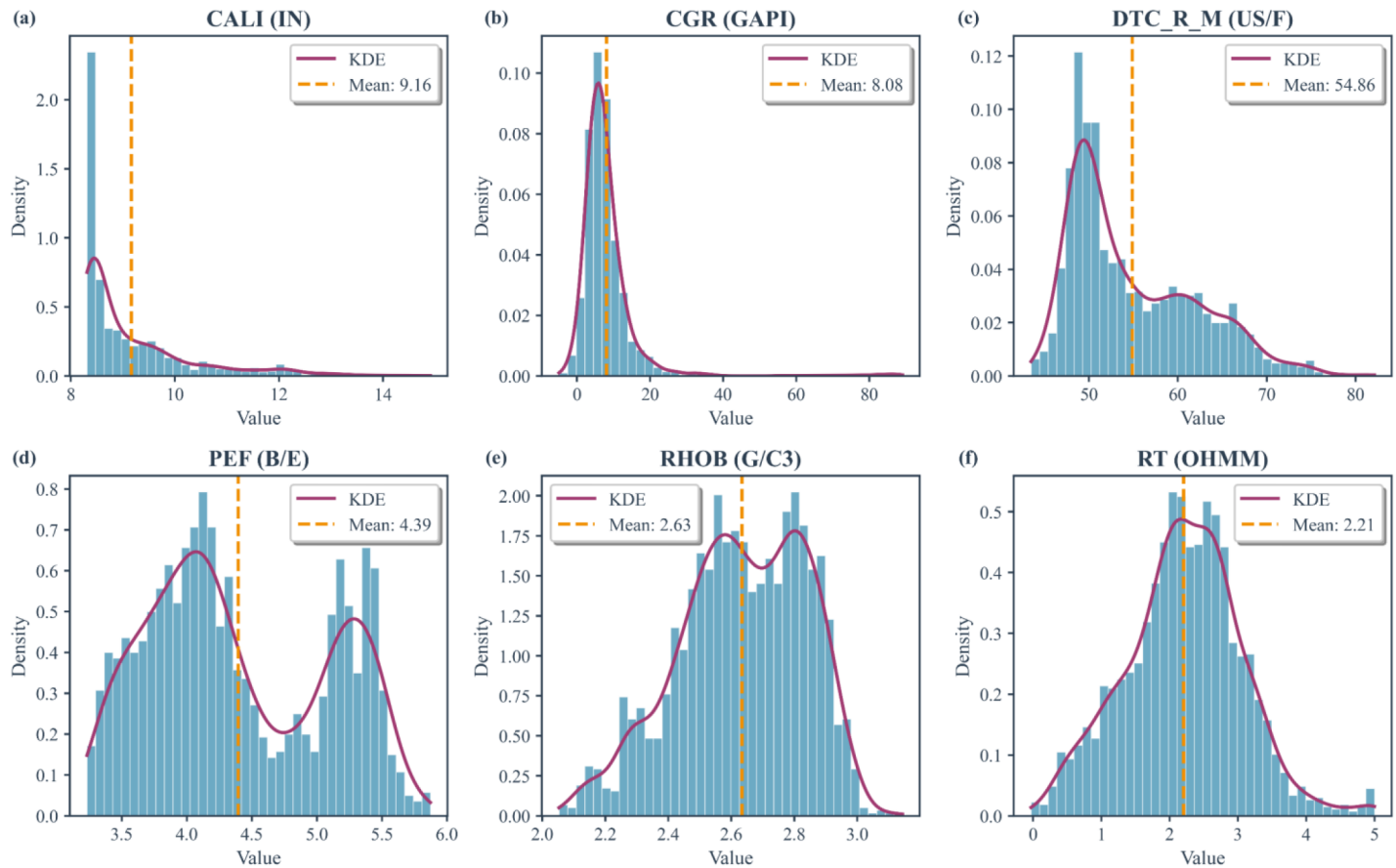


Figure 2—Distribution of open-hole log features with KDE and mean for Well A. Histograms and Kernel Density Estimate (KDE) curves are shown for each feature, including Caliper, CGR, DTC, PEF, RHOB, and RT, with dashed vertical lines indicating their mean values. This figure illustrates the statistical distributions of the input features, highlighting key characteristics such as skewness and the presence of outliers.

The feature correlation matrix in [Figure 3](#) illustrates the pairwise correlation coefficients between all open-hole log features and the target variable, DTS. Each cell in the matrix contains a correlation value ranging from -1 to 1, where values near 1 indicate a strong positive correlation, values near -1 indicate a strong negative correlation, and values around 0 suggest no linear relationship. The matrix is typically visualized using a color scale to highlight highly correlated feature pairs easily. This plot is critical because it helps detect multicollinearity among the predictor variables, which can destabilize the multivariate regression model and make coefficient interpretation challenging. ([Alimohammadi, Rahmanifard and Chen, 2020](#)) High correlations between features may signal redundancy, prompting the removal of certain variables or the use of dimensionality reduction techniques like principal component analysis (PCA) to prevent overfitting and improve model generalizability. Moreover, understanding the correlations between the predictors and DTS guides feature selection, ensuring that the most relevant logs are included in the final model. In the context of the South Pars field, where geological complexity drives non-linear relationships, this matrix provides a foundational understanding of linear dependencies, setting the stage for more advanced analyses.

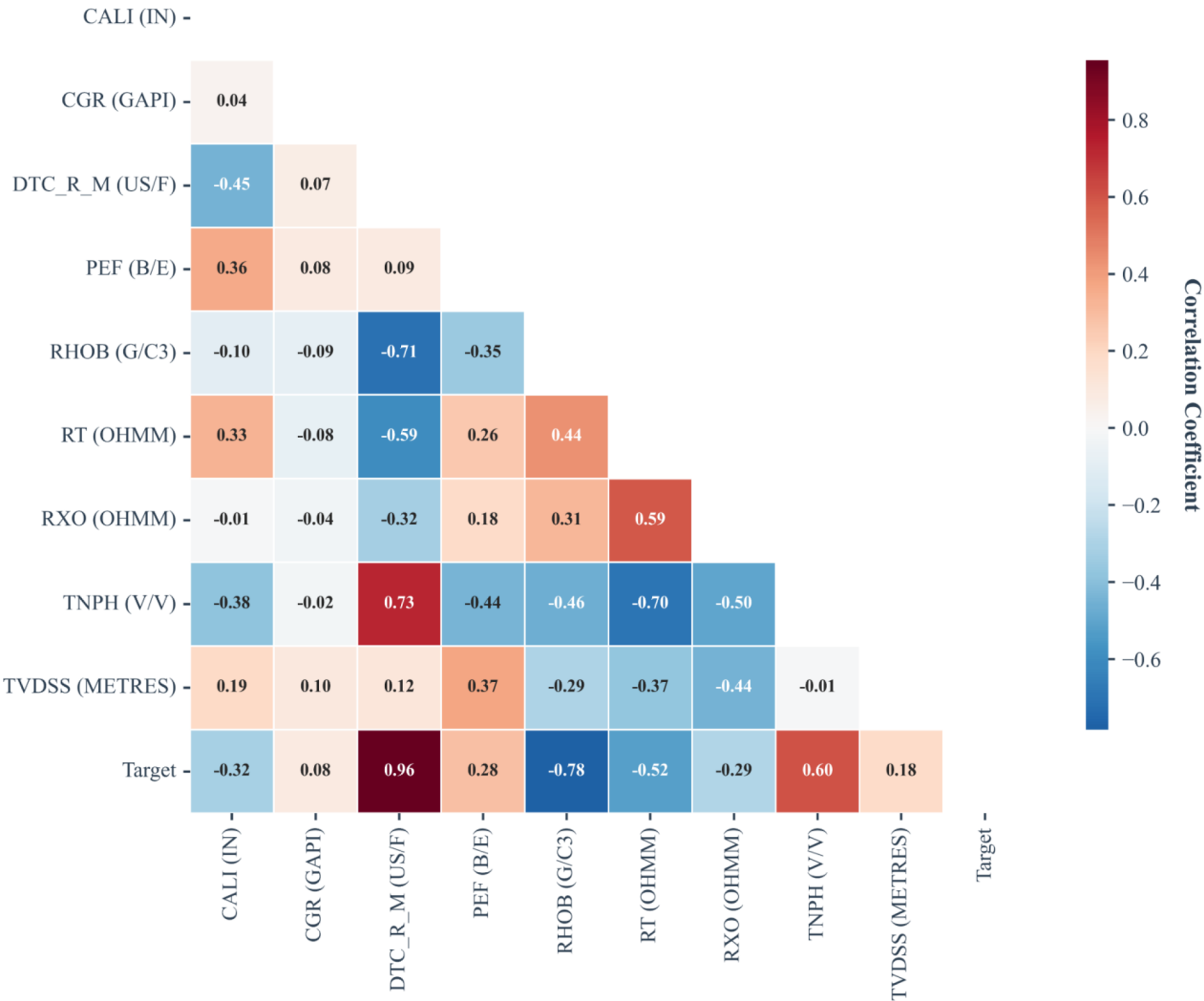


Figure 3—Correlation matrix of open-hole log features for Well A. This matrix presents the pairwise correlation coefficients between all features and the target variable DTS, using a color scale to indicate the strength and direction of linear relationships. It is essential for identifying multicollinearity and guiding feature selection in the multivariate regression analysis.

The pairwise feature relationship plot in Figure 4 consists of a grid of scatter plots, each depicting the relationship between two features from the open-hole log dataset. For every pair of features, a scatter plot is generated, enabling the visual exploration of linear or non-linear associations, clusters, and potential outliers. This comprehensive visualization is vital for understanding how different logs interact and collectively influence the target variable, DTS. In multivariate regression, identifying non-linear relationships can indicate the need for polynomial or interaction terms in the model, thereby improving its predictive accuracy. Additionally, this plot can uncover patterns or anomalies, such as heteroscedasticity or grouping effects tied to different lithological zones, that may not be evident from individual feature distributions or correlation coefficients alone. For instance, in the South Pars reservoir, variations in lithology and diagenesis across the K1 to K4 zones may manifest as distinct clusters in the scatter plots, informing zone-specific modeling approaches. By offering a holistic view of the feature space, the pairwise relationship plot supports informed decisions on feature engineering and model specification, ultimately strengthening the regression analysis's ability to capture the reservoir's complex dynamics.

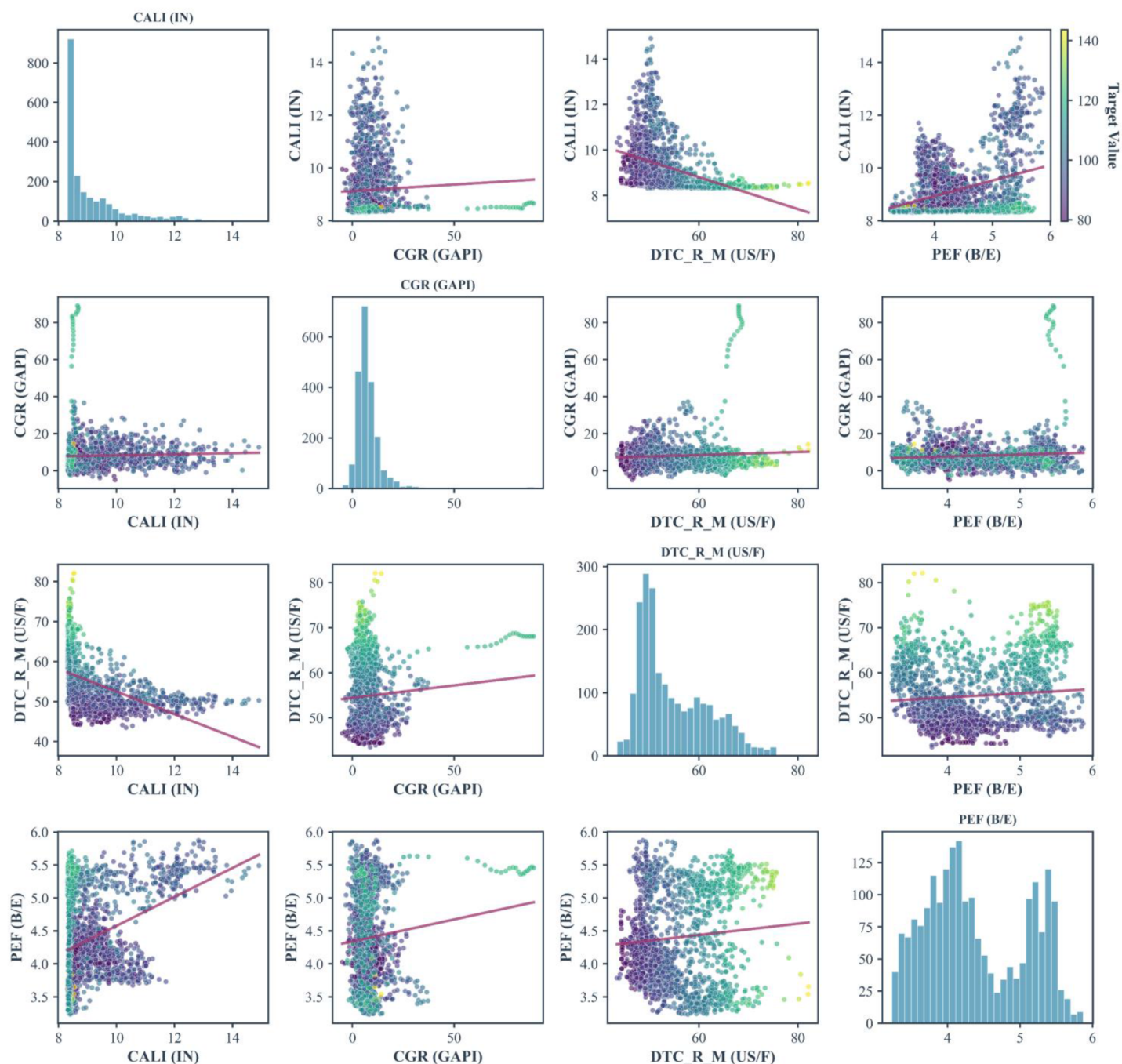


Figure 4—Pairwise relationships between open-hole log features for Well A. A grid of scatter plots depicts the relationships between each pair of features, enabling the visual exploration of linear and non-linear associations, clusters, and potential outliers. This figure is crucial for understanding feature interactions and informing model specification in the regression analysis.

Integration with Geological and Geophysical Interpretation

Although the presented workflow primarily leverages petrophysical logs, the variability of DTS and DTST in carbonate reservoirs is also strongly influenced by geological and geophysical heterogeneities that extend beyond well-scale measurements. Seismic attributes, for instance, capture subsurface variability at the inter-well scale and can provide complementary information on lithology, porosity, and fluid distribution. Acoustic impedance and V_p/V_s ratios derived from pre-stack seismic inversion are particularly relevant, as they directly reflect elastic properties that govern shear- and Stoneley-wave velocities. High V_p/V_s ratios typically signal shaly or fluid-rich facies that correspond to slower DTS/DTST, whereas low

ratios are associated with more rigid, carbonate-rich intervals. Likewise, seismic coherence and curvature attributes are valuable indicators of structural features such as faults and fractures, which can locally alter wave propagation. Stoneley waves, being sensitive to fracture-related permeability, may therefore display systematic anomalies in fractured intervals that can be better captured if such seismic attributes are incorporated into the feature set.

At the scale of stratigraphic modeling, facies distributions and depositional architecture are additional sources of explanatory power. For example, the shallow marine ramp system of the Kangan and Dalan formations produces lateral variations in grain-dominated shoals versus mud-dominated facies. These contrasts are evident in seismic-scale facies models and can inform ML predictions of acoustic properties by conditioning the models to expect slower velocities in mud-rich, high-porosity facies and faster velocities in tight, cemented carbonates. Integration of these geophysical inputs with the conformalized ensemble workflow would therefore extend the predictive reach from well-log scale to field scale, improving inter-well generalization and enhancing relevance for multidisciplinary reservoir studies.

Results and Discussion

This section presents a comprehensive, multi-layered analysis of the modeling results, structured across three distinct predictive scenarios. The analysis begins by establishing the optimal point-prediction models for both Shear (DTS) and Stoneley (DTST) transit times within a single well. It then evaluates the practical challenge of inter-well generalization. Finally, it applies the CQR framework to the best-performing models to generate rigorous uncertainty bounds, which are then interpreted in a geological and engineering context.

Scenario 1: Single-Well DTS Prediction (Well A)

The first scenario focused on establishing the most accurate methodology for predicting DTS within a single, data-rich well (Well A). This involved a rigorous benchmarking of multiple algorithms to select the optimal point predictor.

Comparative Model Performance and Selection. A comprehensive suite of eight ML models was trained and evaluated to predict DTS. This included four tree-based ensemble methods (LightGBM, XGBoost, Random Forest, CatBoost), three regularized linear models (Lasso, Ridge, ElasticNet), and a stacked ensemble model. As Figure 5, the results unequivocally demonstrate the superiority of tree-based ensemble methods for this task.

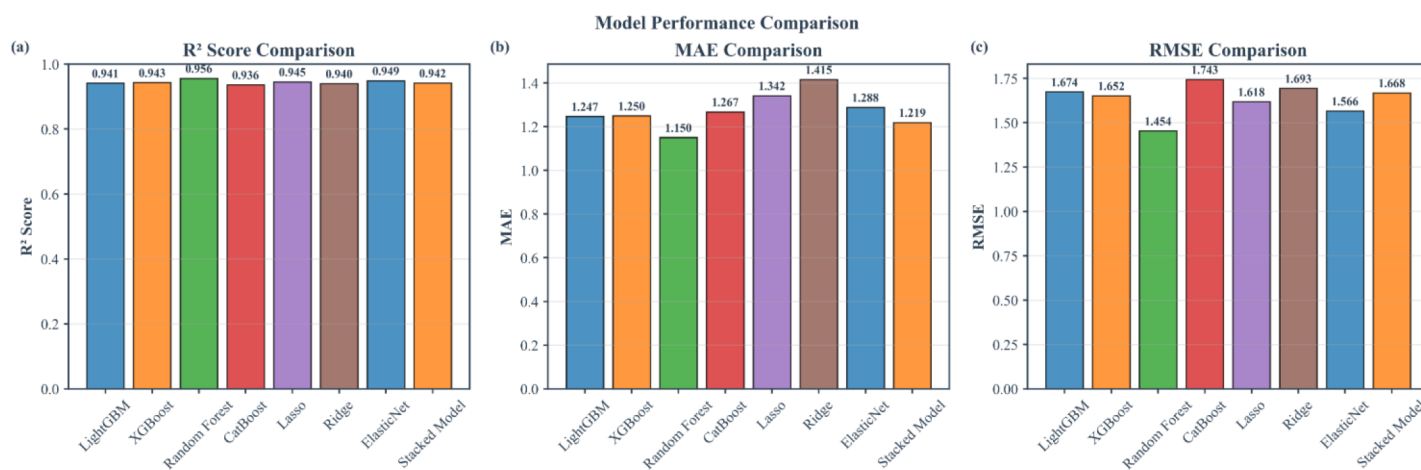


Figure 5—Model Performance Comparison for DTS Prediction. This bar chart displays the R^2 , MAE, and RMSE for all eight evaluated models on the test set, clearly illustrating the superior accuracy of the Random Forest model.

The RF model emerged as the definitive top performer, achieving the highest R^2 of 0.956 and the lowest MAE and RMSE of 1.15 and 1.454, respectively. This confirms the findings in the literature that ensemble methods, particularly RF, are exceptionally well-suited for capturing the complex, non-linear relationships inherent in petrophysical data. The linear models, while computationally efficient, lacked the capacity to model this complexity and performed significantly worse.

Notably, the stacked ensemble, which combined the four best-performing models, did not surpass the best individual model. An analysis of the meta-model weights (XGBoost: 0.856, Ridge: 0.205, Lasso: 0.093, ElasticNet: -0.15) suggests that within a single well where data distributions are relatively consistent, the added complexity of stacking did not provide a benefit over a single, well-tuned, and robust model like Random Forest.

Diagnostic Analysis of the Optimal Model (Random Forest). A deep diagnostic analysis of the winning RF model confirms its robustness and provides petrophysical insights. The cross-plot of predicted versus actual DTS values (Figure 6) shows excellent performance across all data splits, with the test set achieving an R^2 of 0.942 and an MAE of 1.219 $\mu\text{s}/\text{ft}$. This demonstrates strong generalization to unseen data.

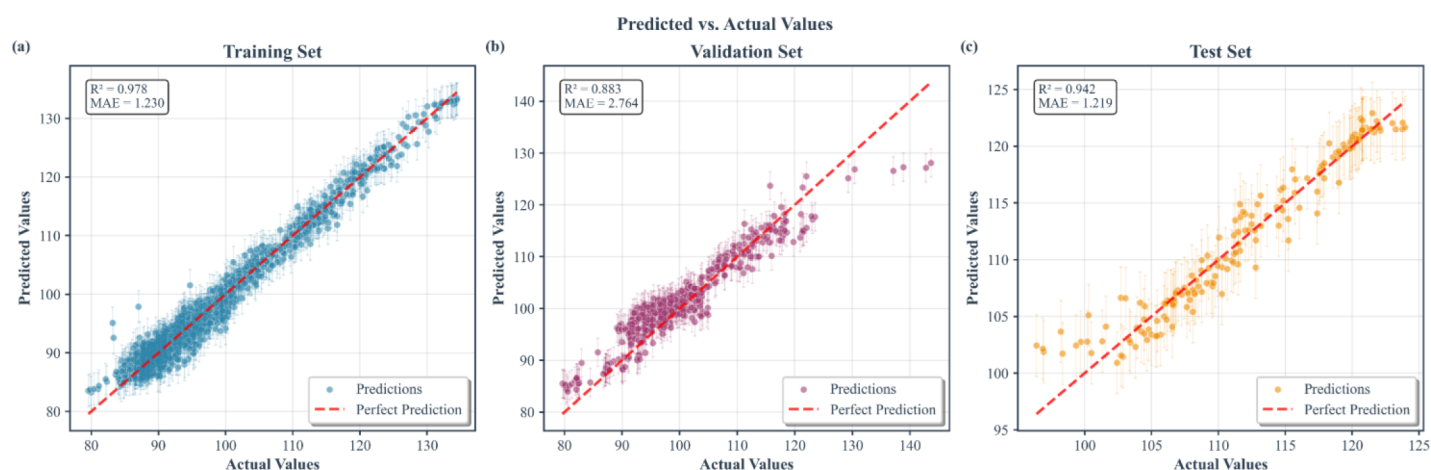


Figure 6—Predicted Vs. Actual DTS Values for the Optimal RF Model. This plot shows a tight correlation between predicted and true DTS values for the training ($R^2=0.978$), validation ($R^2=0.883$), and unseen test ($R^2=0.942$) datasets, confirming the model's high accuracy and generalization capability. Each point is also displayed with its associated prediction interval.

Residual analysis (Figure 7) further validates the model, showing randomly distributed errors centered around zero with no systematic bias. The accompanying Q-Q plots confirm that the residuals approximate a normal distribution, satisfying a key assumption for statistical validity.

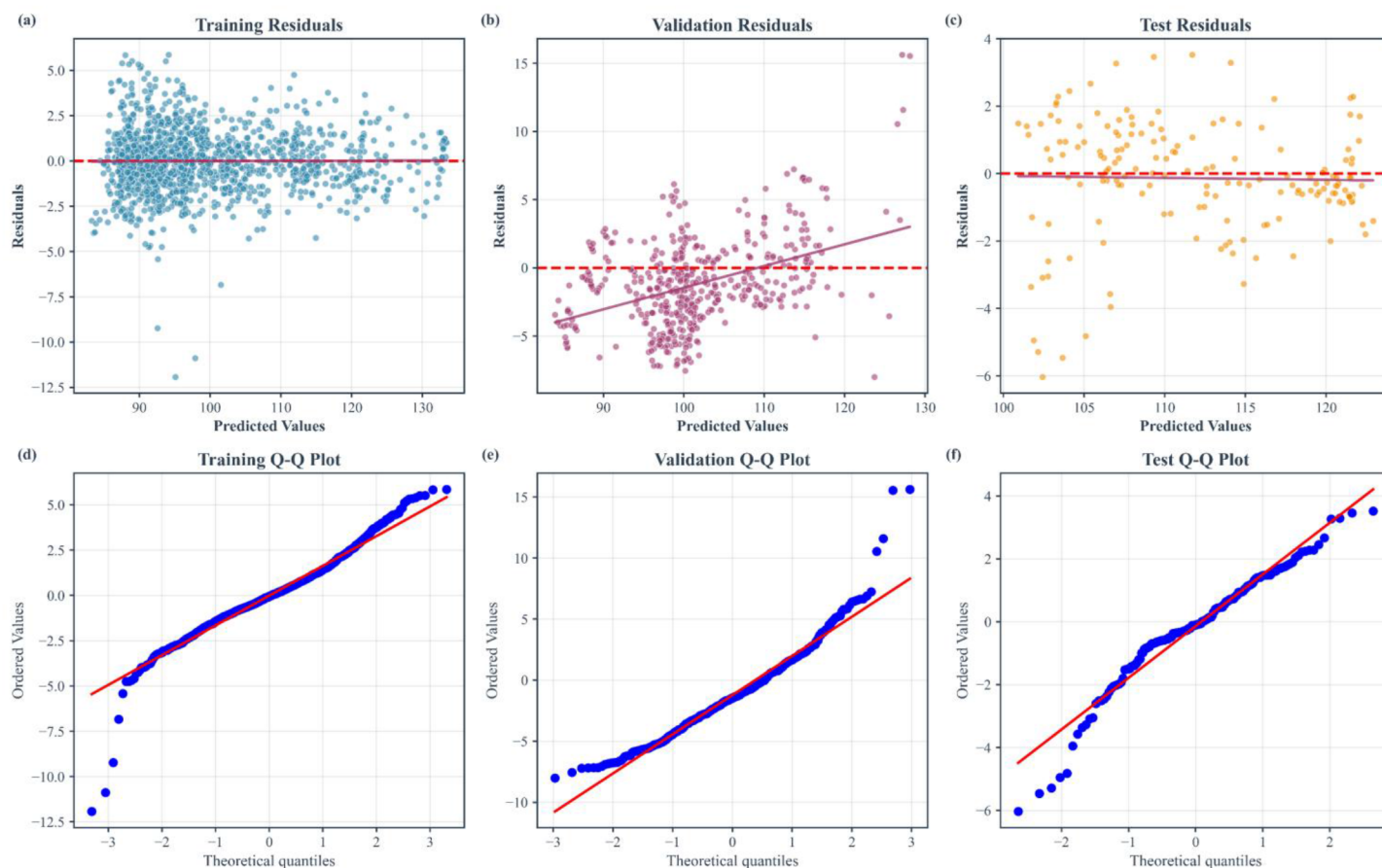


Figure 7—Residual Analysis for the Optimal RF Model. The plots display the distribution of prediction errors (residuals) and Quantile-Quantile (Q-Q) plots for the training, validation, and test sets. The random, unbiased nature of the residuals and their adherence to a normal distribution indicate a well-fitted model.

Feature importance analysis (Figure 8) reveals that the Caliper (CALI) log is the most influential predictor. This is a critical insight, as it underscores the sensitivity of shear wave propagation to borehole conditions. An enlarged or irregular borehole, indicated by the caliper log, can significantly alter the acoustic tool's response, and the model correctly learned to heavily weigh this factor. Conversely, Bulk Density (RHOB) and Photoelectric Factor (PEF) were found to be the least important, suggesting their predictive information was largely redundant or better captured by the combination of other logs.

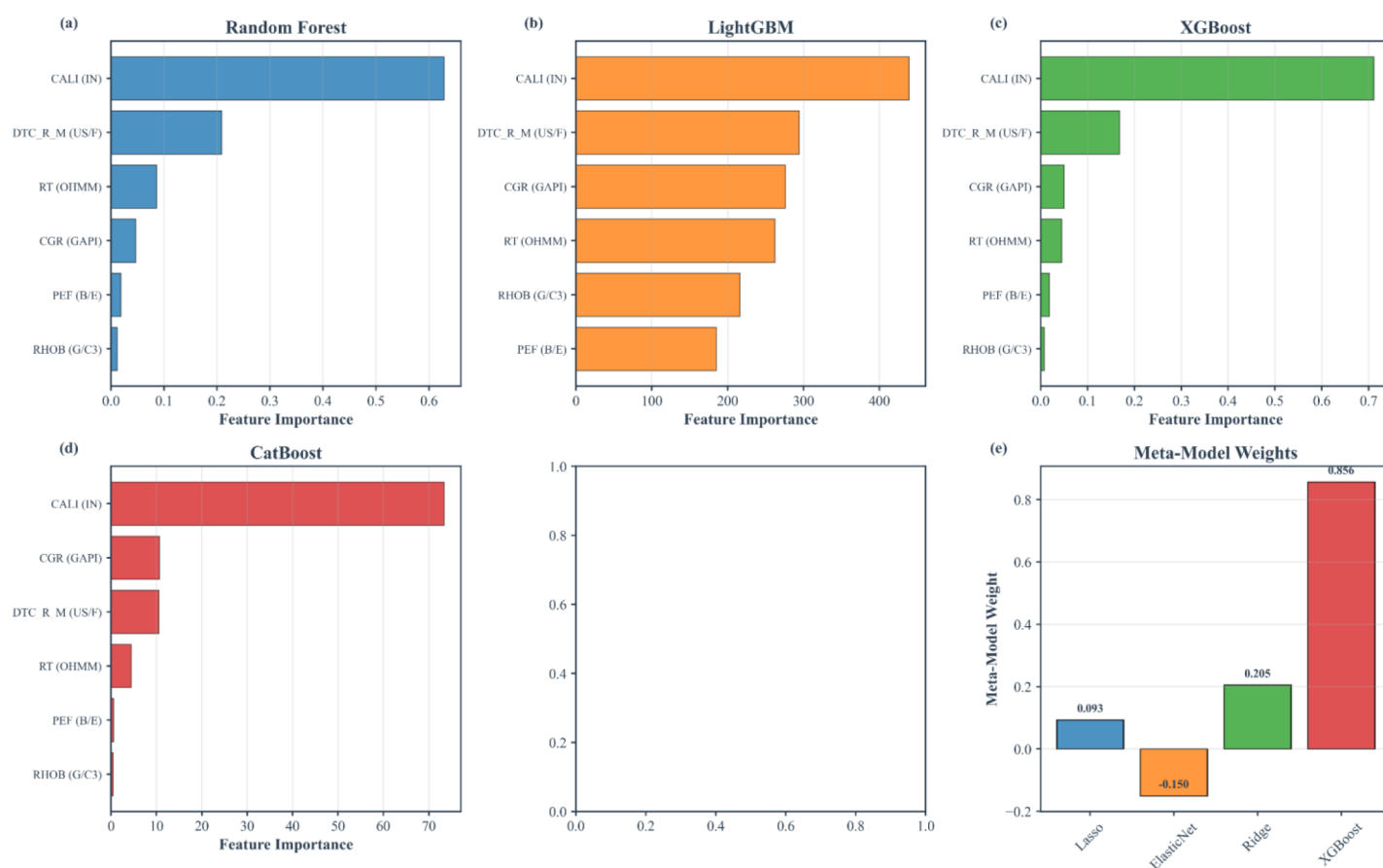


Figure 8—Feature Importance and Meta-Model Weights. The left panel shows the feature importance scores for the tree-based models, highlighting Caliper (CALI) as the most predictive feature for DTS. The bottom right panel displays the weights assigned by the meta-model in the stacked ensemble, showing a heavy reliance on XGBoost.

Scenario 2: Single-Well DTST Prediction (Well A)

To ensure the methodology was robust, the entire evaluation process was repeated for the prediction of Stoneley Transit Time (DTST) in Well A.

Model Performance and Feature Insights. The results for DTST prediction mirrored those for DTS are displayed in Figure 9 and Figure 10, with Random Forest again emerging as the top-performing model, achieving an excellent test set R^2 of 0.936, MAE of 0.648 $\mu\text{s}/\text{ft}$, and RMSE of 0.848 $\mu\text{s}/\text{ft}$. The diagnostic plots for the DTST model were similarly strong, with the test set showing an R^2 of 0.885 and MAE of 0.887 $\mu\text{s}/\text{ft}$ on the predicted Vs. actual plot.

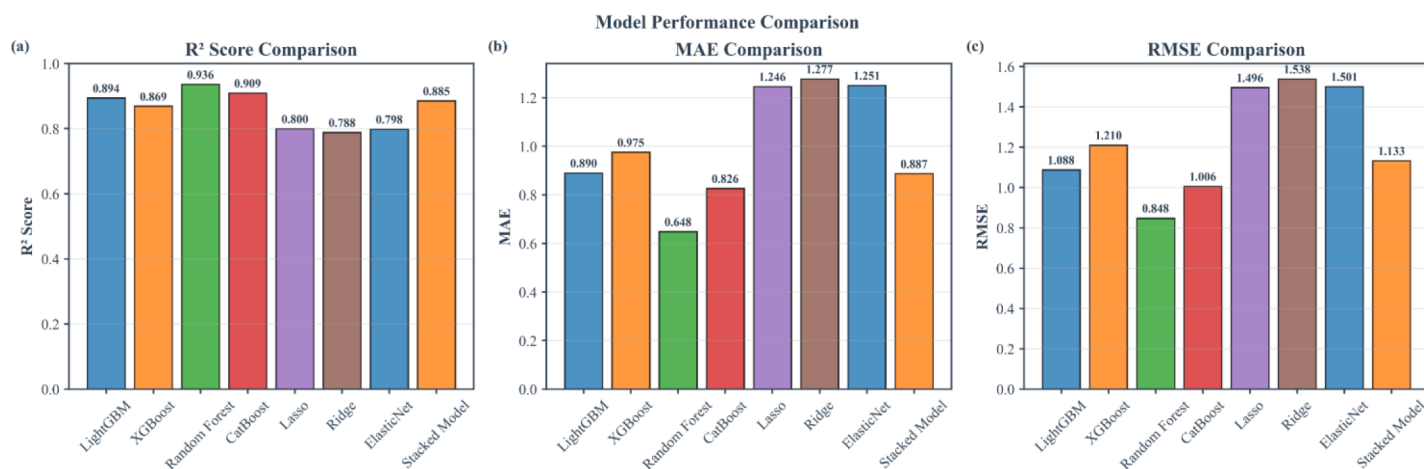


Figure 9—Model Performance Comparison for DTST Prediction. This bar chart displays the R^2 , MAE, and RMSE for all eight evaluated models on the test set, clearly illustrating the superior accuracy of the Random Forest model.

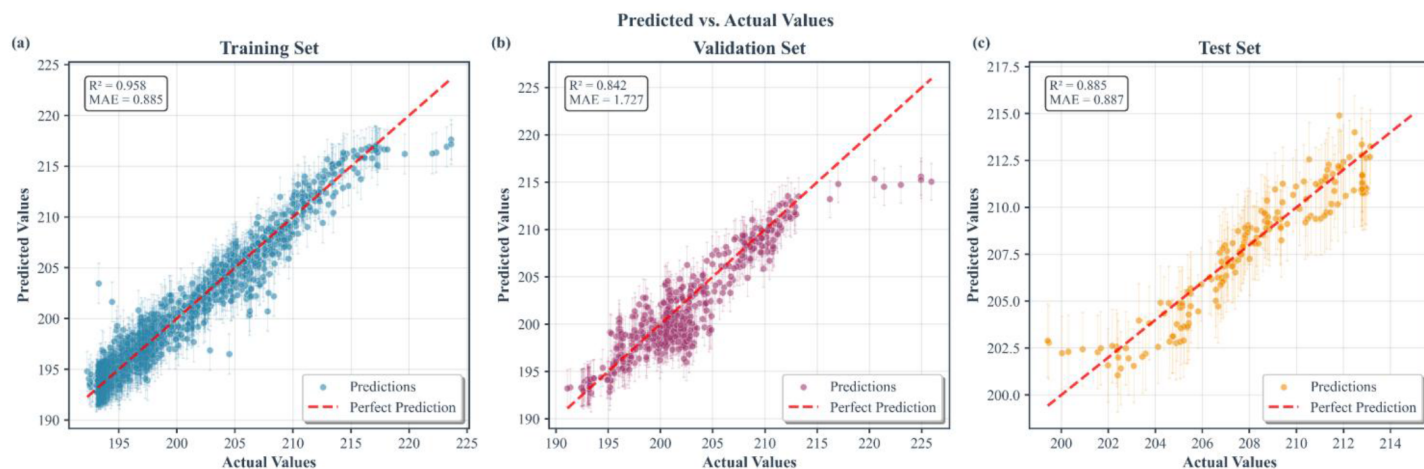


Figure 10—Predicted Vs. Actual DTST Values for the Optimal RF Model. This plot shows a tight correlation between predicted and true DTST values for the training ($R^2=0.958$), validation ($R^2=0.842$), and unseen test ($R^2=0.885$) datasets, confirming the model's high accuracy and generalization capability. Each point is also displayed with its associated prediction interval.

The feature importance analysis for DTST in Figure 11, however, revealed a different, yet petrophysically intuitive, ranking. The most important feature for predicting DTST was overwhelmingly the Compressional Transit Time (DTC). This is physically sound, as the Stoneley wave is a guided wave that travels along the borehole interface, and its velocity is fundamentally linked to the properties of both the borehole fluid and the formation, which are also captured by the compressional wave. The least important features were deep resistivity (RT) and compensated gamma ray (CGR). The stacked model weights for DTST also showed a different combination (CatBoost: 0.978, Ridge: 0.281, Lasso: -0.114, ElasticNet: -0.135), further indicating that the optimal model combination is target-specific.

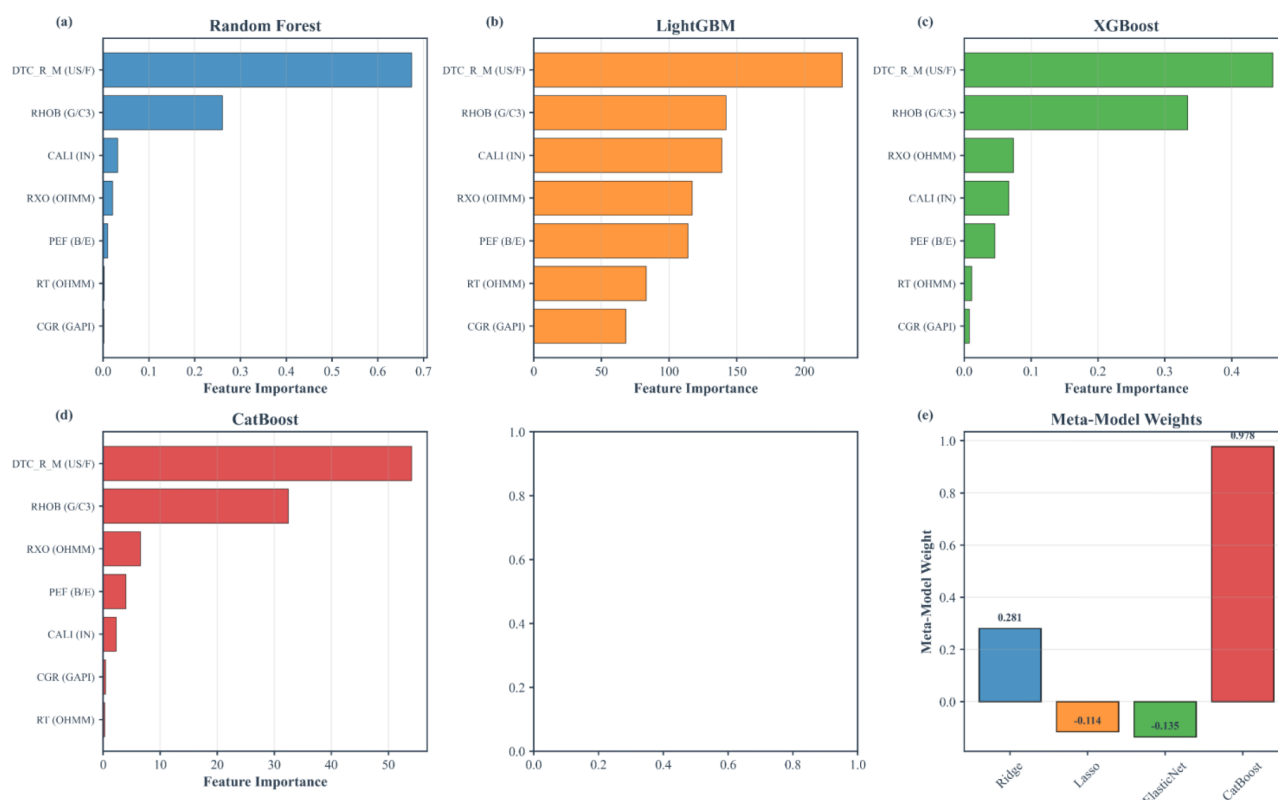


Figure 11—Feature Importance and Meta-Model Weights. The left panel shows the feature importance scores for the tree-based models, highlighting DTC as the most predictive and RT and CGR the least predictive features for DTST. The bottom right panel displays the weights assigned by the meta-model in the stacked ensemble, showing a heavy reliance on CatBoost.

Scenario 3: Inter-Well Generalization for DTST Prediction

The most stringent test of data-driven workflow is its ability to generalize to new wells where the target log is absent. In this scenario, models were trained on data from two wells (Well A and Well B) and tasked with predicting DTST in a third, completely blind well (Well C).

The Power of Ensembling for Robustness. The results of this inter-well test, displayed in Figure 12, revealed a critical finding. In this more challenging generalization task, the Stacked Ensemble model outperformed all individual models, achieving a test R^2 of 0.925, MAE of 1.325 and RMSE of 1.78.

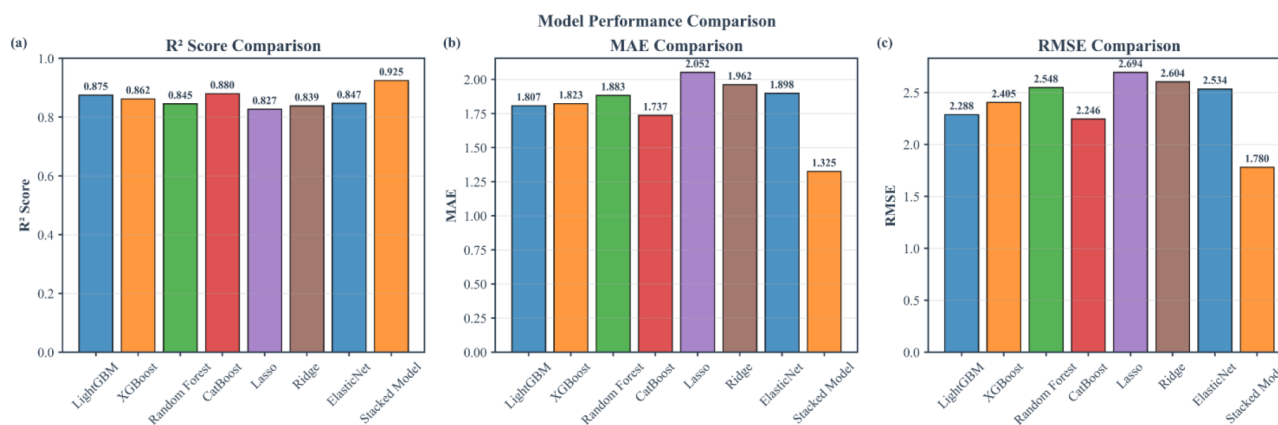


Figure 12—Model Performance Comparison for DTST Prediction with Well C used as test well. This bar chart displays the R^2 , MAE, and RMSE for all eight evaluated models on the test set, clearly illustrating the superior accuracy of the stacked model.

This reversal in performance compared to the single-well scenario is highly significant. It suggests that when predicting on a new well, which will inevitably have subtle geological variations and data distribution shifts compared to the training wells, combining the predictions of several diverse, high-performing models (i.e., stacking) creates a more robust and generalizable final model. Each base model may capture slightly different aspects of the complex petrophysical relationships, and the meta-learner effectively synthesizes these diverse "perspectives" to produce a superior prediction on unseen data. The model's predicted Vs. actual plot for this scenario confirms its high accuracy, with the test set achieving the reported R^2 of 0.925 and MAE of 1.325 $\mu\text{s}/\text{ft}$ as displayed in Figure 13.

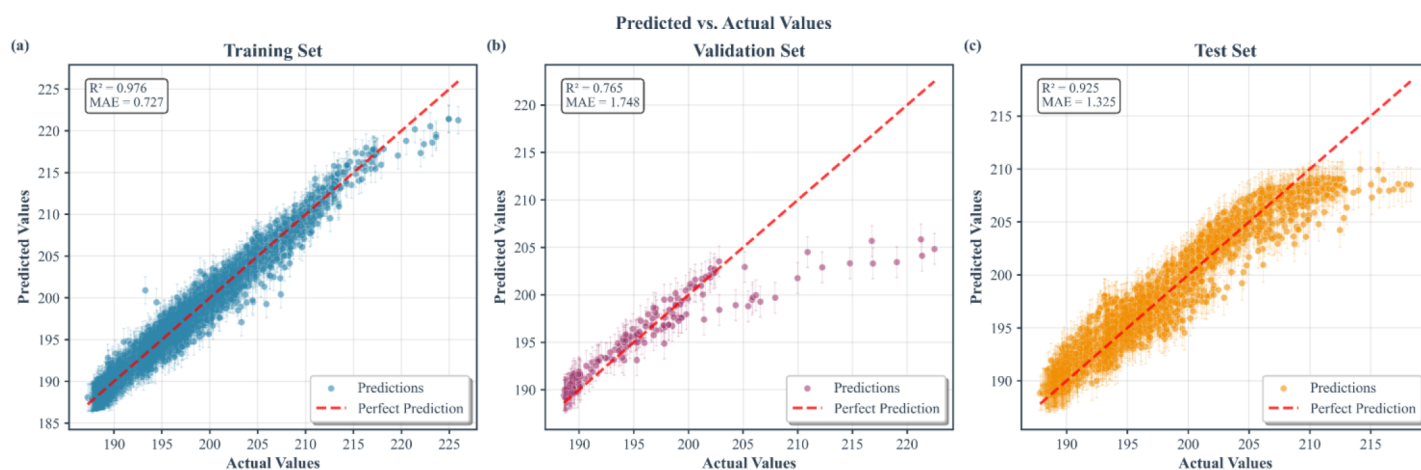


Figure 13—Predicted Vs. Actual DTST Values for the stacked Model. This plot shows a tight correlation between predicted and true DTST values for the training ($R^2=0.976$), validation ($R^2=0.765$), and unseen test ($R^2=0.925$) datasets, confirming the model's high accuracy and generalization capability. Each point is also displayed with its associated prediction interval.

Rigorous Uncertainty Quantification with Conformalized Quantile Regression (CQR)

Having identified the optimal point predictors for each scenario, the CQR framework was applied to generate statistically rigorous, sample-specific 90% prediction intervals. The performance of the CQR method was evaluated to confirm its validity and efficiency (Figure 14 for DTS of Well A, Figure 15 for DTST of Well A, and Figure 16 as DTST of well C used as test data). The coverage calibration plot demonstrates that the actual, empirical coverage of the intervals closely matches the ideal 90% target for single well analysis, providing a strong guarantee of their reliability. The analysis also shows that as the desired confidence level increases, the intervals become wider, and that the coverage-to-width ratio remains high, indicating the intervals are as narrow as possible while maintaining their statistical guarantee. It is worth noting that the coverage calibration plot shows that the empirical coverage of the intervals falls below the target coverage when data from Well A and Well B is used to estimate the DTST of Well C. This indicates a higher level of uncertainty when the test data comes from a different well that is not represented in the training set.

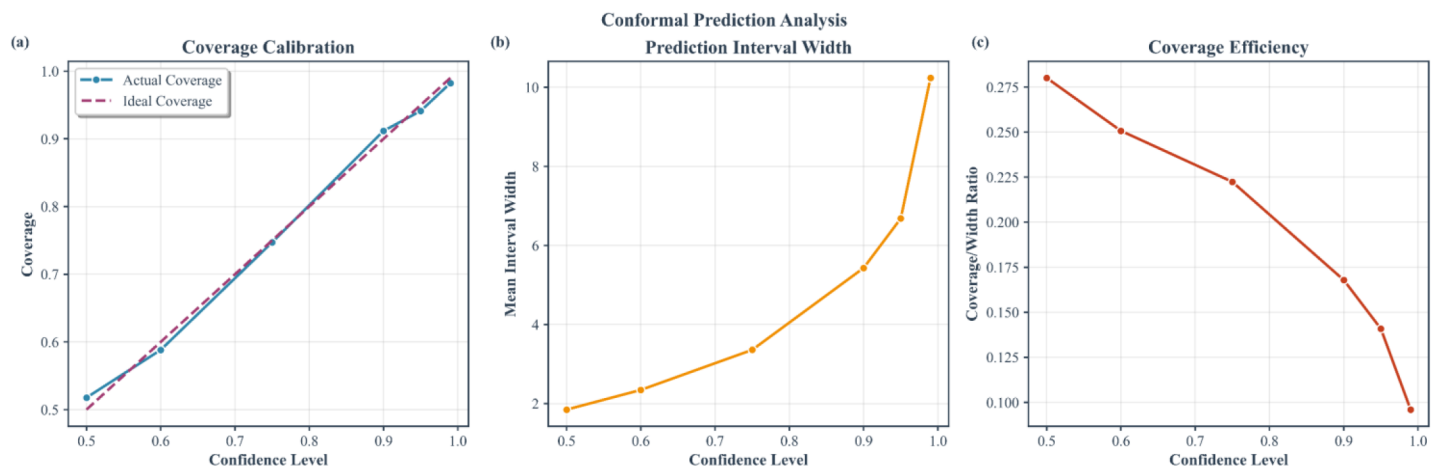


Figure 14—Conformal Prediction Analysis of DTS of well A. These plots validate the CQR framework by showing (a) near-perfect alignment between actual and ideal coverage, (b) a logical increase in interval width with higher confidence levels, and (c) high coverage efficiency, confirming the intervals are both reliable and informative.

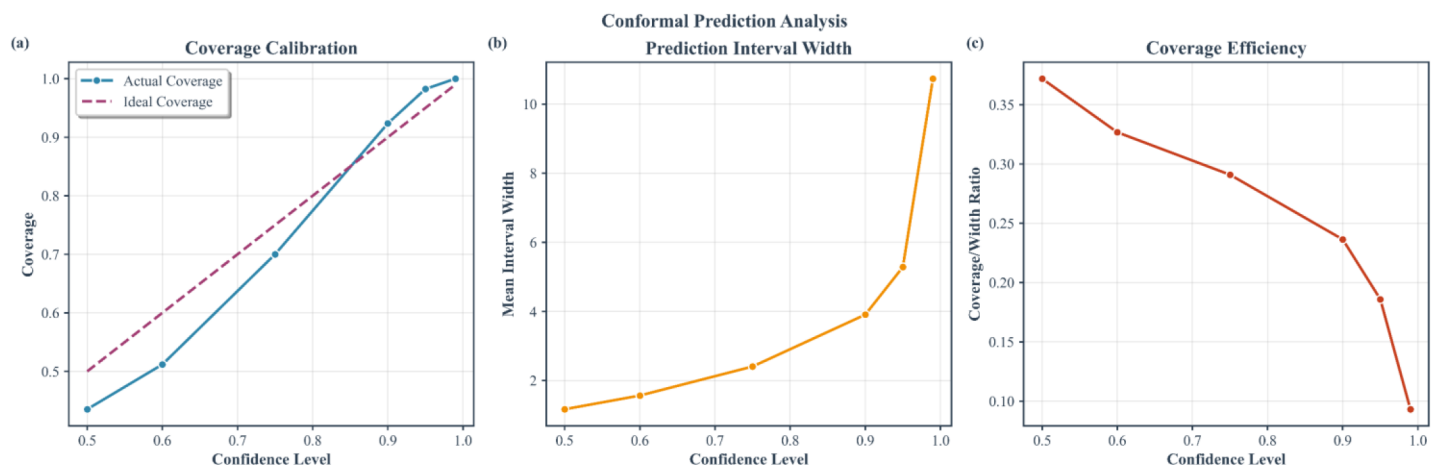


Figure 15—Conformal Prediction Analysis of DTST of well A. These plots validate the CQR framework by showing (a) lower alignment between actual and ideal coverage compared to DTS, (b) a logical increase in interval width with higher confidence levels, and (c) high coverage efficiency, confirming the intervals are both reliable and informative.

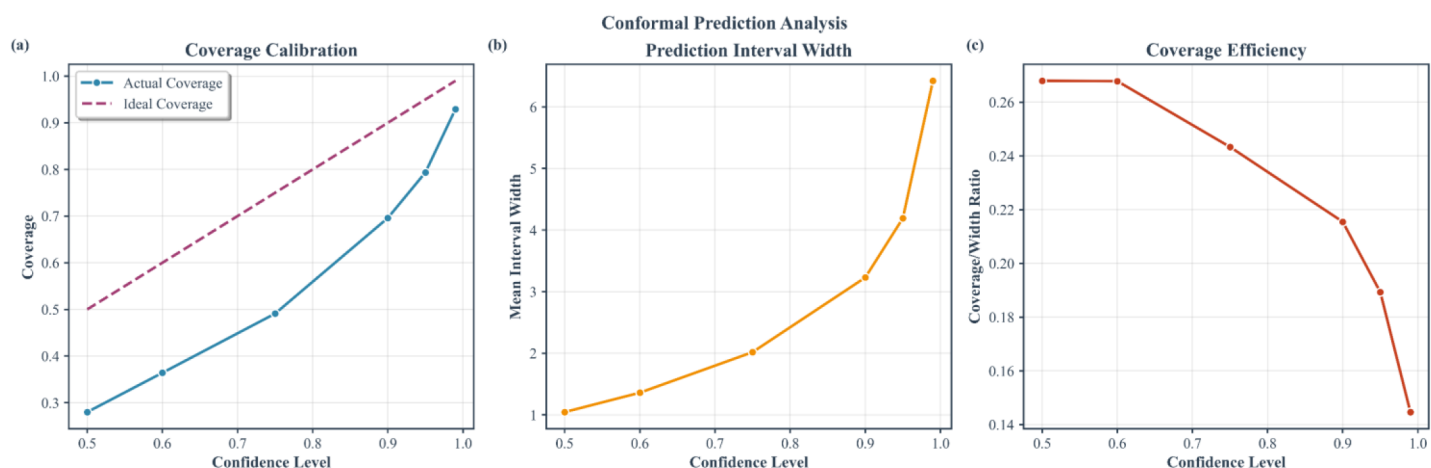


Figure 16—Conformal Prediction Analysis of DTST of well C. These plots validate the CQR framework by showing (a) lower alignment between actual and ideal coverage compared to DTS, (b) a logical increase in interval width with higher confidence levels, and (c) high coverage efficiency, confirming the intervals are both reliable and informative.

The final output, visualized in a depth log plot (Figure 17 and Figure 18), shows the highly accurate point predictions from the optimal model tracking the true log, enveloped by the 90% prediction interval band. This band provides an intuitive and quantitatively reliable measure of uncertainty at every depth point.

To further evaluate the predictive performance of the proposed methodology, results were examined within the stratigraphic framework of the K1–K4 reservoir zones. These zones, which are distinguished by their depositional textures and diagenetic histories, provide a natural template to assess whether model accuracy and uncertainty behave consistently with geological expectations. Depth-wise visualization of predictions, overlaid on actual measurements and annotated by reservoir zone, reveals a clear pattern.

In the K1 and K3 intervals, which are dominated by anhydrite and exhibit limited porosity, the predicted DTS and DTST values show narrow prediction intervals and low residual error. This behavior is expected, as homogeneous, low-porosity lithologies present relatively stable acoustic properties with limited heterogeneity. Conversely, in the K2 and K4 intervals, both primary reservoir zones composed of dolomitized limestone with significant secondary porosity, the models exhibit wider prediction intervals and larger variance in residuals. This reflects the inherent heterogeneity of these intervals, where diagenetic dissolution, vuggy porosity, and occasional fracturing introduce localized acoustic variability that challenges deterministic prediction.

By mapping uncertainty behavior to stratigraphy, the workflow not only quantifies the reliability of predictions but also highlights zones that merit closer attention in reservoir evaluation. Narrow confidence bands in anhydritic K1 and K3 suggest predictions can be used with high confidence in static and dynamic models, whereas the broader bands in K2 and K4 flag intervals where additional measurements or conservative engineering assumptions are advisable. Such stratigraphic visualization provides interpretable, zone-specific diagnostics that bridge ML predictions with geological understanding.

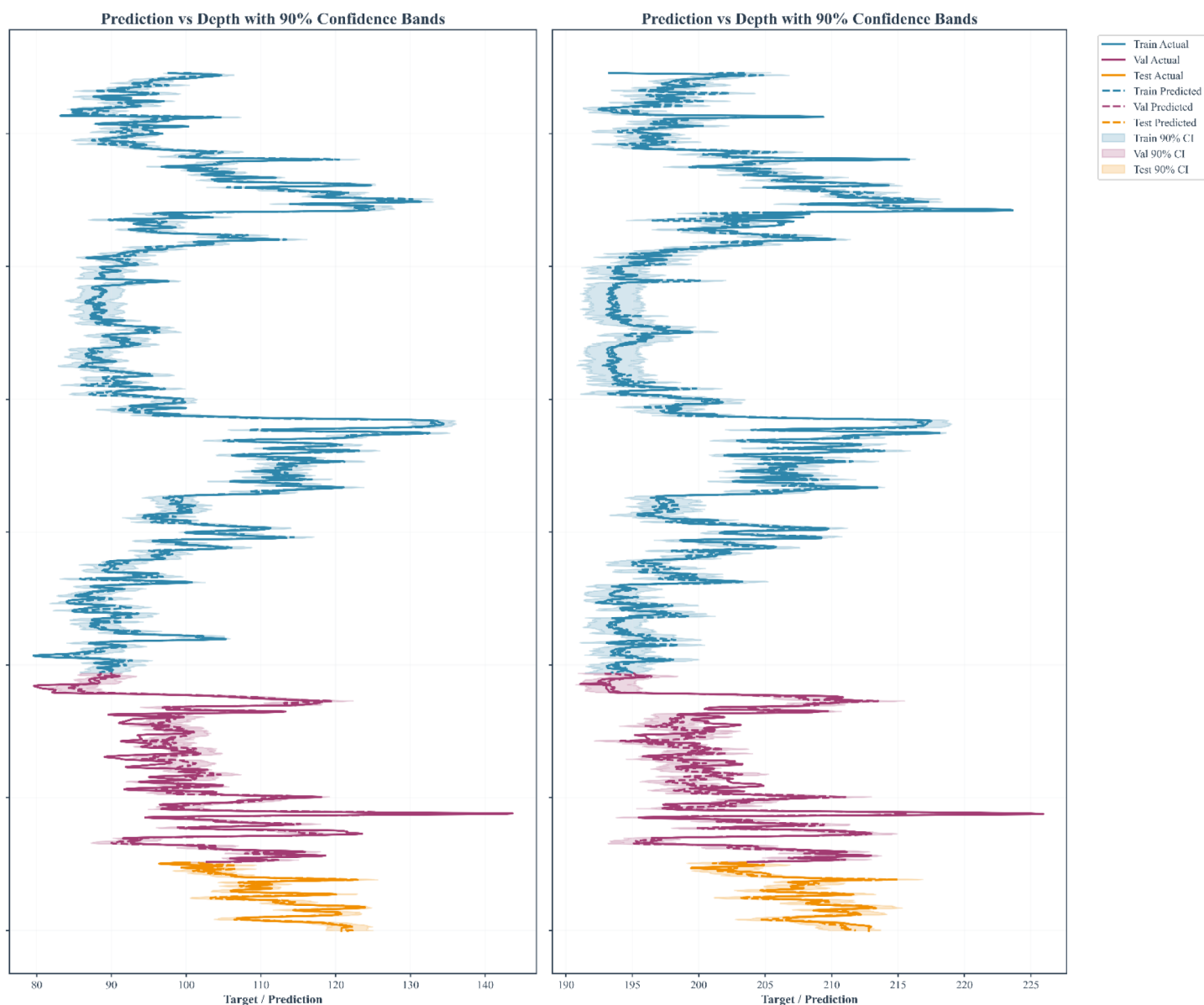


Figure 17—Depth log with conformal prediction intervals (left: DTS of Well A; right: DTST of Well B). This plot shows the actual and predicted DTS and DTST curves versus depth for the training, validation, and test sets. The shaded region represents the 90% prediction interval, which successfully contains the true log for most points, visually confirming the validity of the CQR framework.

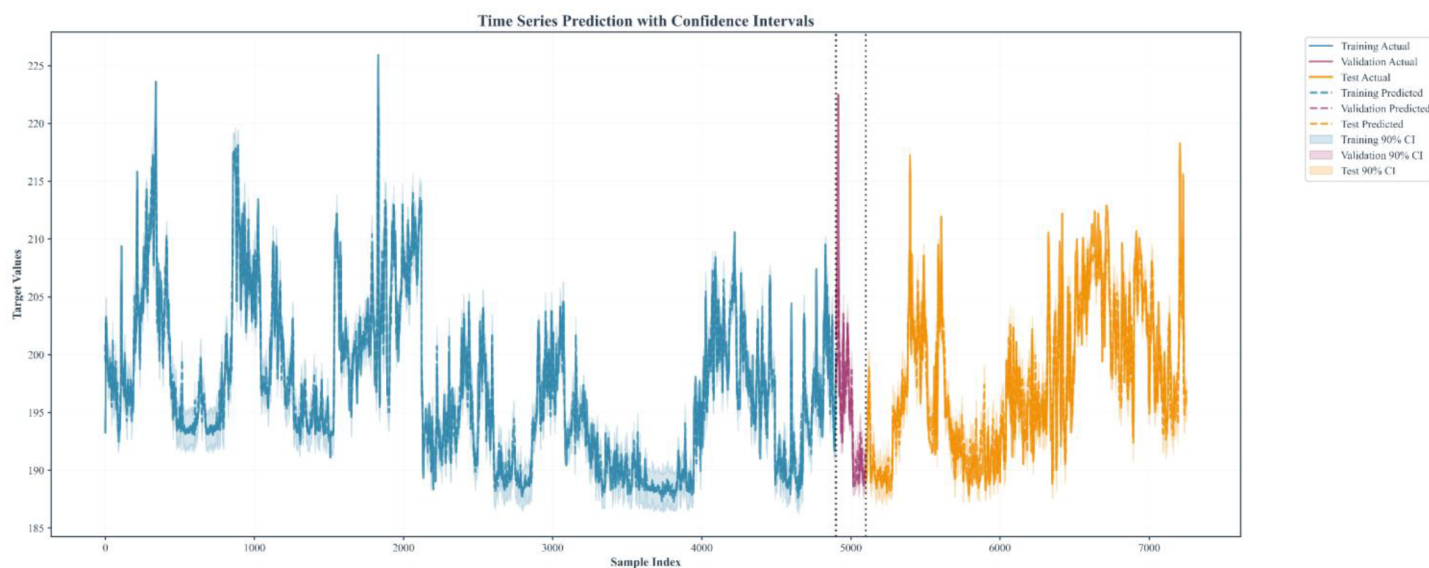


Figure 18—Index log with conformal prediction. This plot shows the actual and predicted DTST curves across the training, validation, and test sets where the data of Wells A and B is used for training and validation and Well C for testing. The shaded region represents the 90% prediction interval, which successfully contains the true log for most points, visually confirming the validity of the CQR framework. Since two wells are used for training and validation and one well for testing, the plots are shown against index rather than depth.

Physical Plausibility and Lithological Constraints

While machine learning models excel at capturing complex statistical relationships, their credibility ultimately rests on whether outputs remain consistent with established rock physics and petrophysical trends. To assess this, the feature importance analysis and depth-wise predictions were interpreted against known lithological and porosity variations in the Kangan and Dalan formations. The results confirm that the models respect these physical trends.

For shear-wave transit time (DTS), predictions showed a strong dependence on caliper and gamma-ray logs, which are proxies for borehole conditions and clay content, respectively. This aligns with the known sensitivity of shear-wave propagation to borehole enlargement and to the rigidity contrast between clean carbonates and clay-rich intervals. Importantly, predicted DTS values systematically increased in clay-rich and porous facies, in agreement with the inverse correlation between shear velocity and porosity observed in carbonate reservoirs. Similarly, for Stoneley-wave transit time (DTST), compressional transit time (DTC) emerged as the dominant predictor, reflecting the physical linkage between Stoneley propagation and the elastic properties of both the borehole fluid and the formation.

These findings reinforce that the models did not generate spurious statistical relationships but instead extracted physically meaningful dependencies. The predictions respect the expected velocity–porosity and velocity–lithology relationships across the reservoir. Moreover, the widening of prediction intervals in zones with higher porosity variability (e.g., vuggy dolomitized K2) provides additional confidence that the conformalized framework captures geological complexity rather than overfitting noise. Taken together, these observations strengthen the physical plausibility of the workflow and support its interpretability within geoscientific and engineering contexts.

Practical Implications of Predictive Uncertainty

The adoption of conformalized quantile regression provides more than statistical rigor; it introduces a practical tool for decision-making in reservoir characterization and engineering design. The key output, sample-specific prediction intervals, directly informs risk assessment at the depth of interest. Narrow prediction intervals indicate high certainty and can be used with confidence in geomechanical calculations,

such as elastic modulus estimation for wellbore stability or fracture design. In such cases, the ML-based predictions may replace missing acoustic logs without requiring further validation.

In contrast, wide prediction intervals serve as explicit signals of uncertainty and should be interpreted as red flags. For example, a broad interval in DTS within a fracture-prone dolomitic zone may suggest the need for additional sonic logging or core sampling to constrain mechanical properties before finalizing a drilling or completion plan. Similarly, wide DTST intervals in heterogeneous intervals may guide reservoir engineers to adopt conservative permeability or fracture-height growth assumptions in dynamic models. In exploration or appraisal wells, such uncertainty information can also help prioritize which zones merit advanced data acquisition, such as dipole sonic logging, versus zones where predictions are sufficiently constrained by existing measurements.

By communicating uncertainty explicitly, the workflow reframes ML predictions from being deterministic black boxes into actionable probabilistic tools. Engineers and geoscientists can move beyond asking "What is the predicted value?" to asking "How certain is the prediction, and what does this mean for operational risk?" In this way, prediction intervals become more than statistical diagnostics, they act as decision-support metrics that can guide data acquisition strategies, risk mitigation planning, and resource allocation across multidisciplinary teams.

Limitations and Future Directions

While the proposed CQR-based workflow demonstrates significant advantages, it is important to acknowledge its limitations. The primary theoretical assumption of conformal prediction is data exchangeability. Well-log data is inherently sequential, with spatial correlation between adjacent depth points, which is a technical violation of this assumption. However, recent theoretical and empirical work has shown that split conformal methods are surprisingly robust to moderate deviations from exchangeability, such as in time-series and spatiotemporal data. The valid coverage achieved in this study supports this finding of practical robustness.

Future research could explore several promising avenues. First, the application of conformal prediction methods specifically designed for sequential, or time-series data could provide even tighter theoretical guarantees by explicitly modeling the spatial dependencies in well logs. Second, the development of custom, asymmetric nonconformity scores could tailor the prediction intervals to specific geomechanical risks, where under-prediction of rock strength may have more severe consequences than over-prediction. Finally, integrating the CQR framework with deep learning models like LSTMs or CNNs could combine the feature extraction power of deep learning with the rigorous UQ of conformal prediction.

Conclusion

This paper has addressed the critical credibility gap in the data-driven synthesis of acoustic wave velocities by moving beyond point predictions to a framework of rigorous uncertainty quantification. The study demonstrates a comprehensive workflow that combines state-of-the-art ensemble machine learning models with Conformalized Quantile Regression (CQR) to generate statistically valid, sample-specific prediction intervals for shear and Stoneley wave velocities.

Applied to the complex and heterogeneous carbonate reservoir of the South Pars gas field, the key findings are:

1. **Ensemble Machine Learning Methods Demonstrate Superior Predictive Accuracy, but the Optimal Model Choice is Task-Dependent.**
 - a. The results unequivocally confirm that tree-based ensemble learning methods are exceptionally well-suited for capturing the complex, non-linear relationships inherent in petrophysical log data. For both Shear Transit Time (DTS) and Stoneley Transit Time (DTST) prediction, ensemble

models, specifically Random Forest (RF), XGBoost, LightGBM, CatBoost, and a Stacked Ensemble, consistently and significantly outperformed conventional regularized linear models (Lasso, Ridge, ElasticNet). This finding aligns with a growing body of literature that establishes ensemble techniques as the state-of-the-art for tabular data regression tasks in petrophysics and beyond.

- b. However, a more sophisticated and practically significant conclusion emerged from the comparison of different prediction scenarios. For single-well prediction, where the model is trained and tested on data from the same well, a well-tuned RF model delivered the highest accuracy, achieving a test set R^2 of 0.956 for DTS. This scenario is analogous to a log reconstruction or repair task, where the model operates within a relatively stable data distribution. In contrast, for the more challenging and common task of inter-well generalization, training on data from multiple wells and predicting on a completely blind well, the Stacked Ensemble model proved to be the most robust and accurate performer, achieving a test set R^2 of 0.925 for DTST. This reversal in performance is critical. It suggests that when a model must extrapolate to a new geological setting with inevitable shifts in data distributions, combining the predictions of several diverse, high-performing models through stacking creates a more resilient and generalizable final predictor. The meta-learner in the stacked model effectively synthesizes the varied "perspectives" of its base models, mitigating the risk of failure that a single model might face when encountering slightly different petrophysical relationships in the new well. This provides an actionable guideline for practitioners: the optimal model architecture is not universal but is contingent on the specific engineering application, demanding a more robust ensembling strategy for extrapolation tasks compared to interpolation tasks.

2. The Models Learned Petrophysically Sound and Target-Specific Drivers for Acoustic Wave Velocities.

- a. A key validation of the data-driven approach comes from the feature importance analysis, which revealed that the models learned physically meaningful and distinct drivers for the different acoustic wave phenomena. For the prediction of DTS, the Caliper (CALI) log was identified as the most influential feature. This is a petrophysically sound finding, as the propagation and measurement of shear waves are highly sensitive to the acoustic coupling between the logging tool and the formation. An enlarged, washed-out, or irregular borehole, as indicated by the caliper, can significantly degrade this coupling and alter the tool's response. The model's ability to identify and heavily weigh this factor demonstrates that it is not merely correlating abstract numbers but has captured a fundamental aspect of the measurement physics.
- b. Conversely, the feature importance analysis for DTST prediction revealed that the Compressional Transit Time (DTC) was the overwhelmingly dominant predictor. This is also physically intuitive. The Stoneley wave is a type of guided wave that propagates along the interface between the borehole fluid and the formation. Its velocity is therefore fundamentally linked to the elastic properties of both the formation rock and the borehole fluid, both of which are directly characterized by the compressional wave measurement. The fact that the models identified different primary drivers for DTS and DTST serves as strong evidence against the "black box" criticism often leveled at ML models. It confirms that the trained models have successfully extracted distinct, physically interpretable relationships from the input data, thereby increasing confidence in their predictive capabilities.

3. Conformalized Quantile Regression Provides a Validated and Practical Solution to the Machine Learning Credibility Gap in Geoscience.

- a. The central contribution of this work is the successful implementation and validation of a framework that addresses the "credibility gap" between high model accuracy and the need for reliable predictions in high-stakes applications. The study demonstrated that by augmenting high-performance point predictors with Conformalized Quantile Regression (CQR), it is possible to generate adaptive, sample-specific prediction intervals that are both statistically rigorous and practically informative. The CQR framework, which combines the efficiency of quantile regression with the distribution-free guarantees of conformal prediction, produced 90% prediction intervals for both DTS and DTST.
- b. Crucially, the validity of these intervals was confirmed through evaluation on a held-out test set. The empirical coverage, the actual percentage of true log values that fell within their corresponding prediction intervals, was shown to closely match the nominal 90% target level. This result is significant because it was achieved without making any assumptions about the underlying distribution of the data or the model's prediction errors, a key advantage of the conformal prediction methodology. The final log plots, which show the narrow, adaptive uncertainty bands tightly enveloping the true log measurements, provide a clear visual confirmation of the framework's effectiveness. This work, therefore, presents a validated, practical workflow that moves beyond unreliable, deterministic point estimates. It delivers trustworthy, probabilistic forecasts that provide engineers and geoscientists with a quantitative measure of confidence for each prediction, directly enabling quantitative risk assessment and fostering greater trust in the application of machine learning for reservoir characterization.

Nomenclature

Symbol	Description
V_p	Compressional wave velocity
V_s	Shear wave velocity
V_{st}	Stoneley wave velocity
DTS	Shear sonic transit time
DTST	Stoneley sonic transit time
DTC	Compressional sonic transit time
ρ_b	Bulk density
E	Young's Modulus
ν	Poisson's Ratio
μ	Shear Modulus
K	Bulk Modulus
ML	Machine Learning
DL	Deep Learning
UQ	Uncertainty Quantification
CP	Conformal Prediction
SCP	Split Conformal Prediction
CQR	Conformalized Quantile Regression
RF	Random Forest
XGBoost	Extreme Gradient Boosting
GR	Gamma Ray
RHOB	Bulk Density
NPHI	Neutron Porosity
LLD	Laterolog Deep Resistivity
CALI	Caliper

PEF	Photoelectric Factor
α	Miscoverage rate
$s(x,y)$	Nonconformity score
$q^{\wedge}$	Quantile of nonconformity scores
R^2	Coefficient of determination
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error

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